

The value of convexity: a theoretical and empirical investigation

RICCARDO REBONATO^{*†} and VLADISLAV PUTYATIN[‡]

[†]EDHEC, London, UK

[‡]Morgan Stanley, London, UK

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We explore from a theoretical and an empirical perspective the value of convexity in the US Treasury market. We present a quasi-model-agnostic approach that is rooted in the existence of *some* affine model capable of recovering with good accuracy the market yield curve and covariance matrix. As we show, at least one such model exists, and this is all we require for our results to hold. We show that, as a consequence, the theoretical ‘value of convexity’ purely depends on observable features of the yield curve, and on statistically determinable yield volatilities. We then address the question of whether the theoretical convexity is indeed correctly reflected in the shape of the yield curve. We present empirical results about the predictive power of a strategy based on the discrepancies between the theoretical and the predicted value of convexity. By looking at 30 years of data, we find that neither the strategy of being systematically long or short convexity (and immunized against ‘level’ and ‘slope’ risk) would have been profitable. However, a conditional strategy that looks at the difference between the ‘implied’ and the statistically estimated value of convexity would have identified extended periods during which the proposed approach would have delivered attractive Sharpe Ratios.

Keywords: Asset pricing; Interest rate modelling; Interest rate derivatives; Interest rates; Convexity; Affine models; Term structure models

JEL Classification: G0, G1, G13

1. Introduction

It is customary to model the yield curve by decomposing its shape into a component related to expectations, a risk-premium contribution and a term from convexity. While this is in principle agreed, in practice far lesser attention has been devoted to the study of the third contributor (convexity) than to the other two—or, for that matter, even than to more esoteric factors or effects, such as liquidity (see, e.g. Fontaine and Garcia 2012) or even to the effect of the zero bound (see, e.g. Wu and Xia 2014).

In the context of yield-curve modelling, convexity is a direct consequence of the non-linear relationship between prices and yields (and, of course, of Jensen’s inequality). In a nutshell, if we allow for symmetric up and down movements for the yield of a discount bond, the price of the bond will increase more in correspondence of the yield fall than it will decrease for the yield rise. This is a valuable feature for the holder of the bond, who will therefore be willing to pay for this feature (bid up the price of the bond), thereby lowering its yield. As we shall see in detail, by how much the bond will be bid up (and its yield correspondingly lowered) depends on the investors’ views of the future yield volatility.

An early classic paper by Brown and Schaefer (2000) discussed the qualitative effect of convexity on the long end of the yield curve, and provided some semi-quantitative estimates of the magnitude of the effect. However, we are not aware of a quantitative study of whether convexity is ‘fairly’ priced in the most liquid government bond market, i.e. Treasuries. This is surprising, because the ‘value of convexity’ (i.e. the difference between the market yields and the yields that would prevail in the absence of convexity) is substantial, as figure 1 clearly shows.

Of course, speaking of ‘fairness’ makes reference to a modelling approach, and, probably herein lies the problem: in studies of this type, one always tests a joint hypothesis, that the convexity is correctly priced *and* the model correctly specified. Given the great uncertainty in model specification, any claim that the shape of the yield curve does not ‘fairly’ reflect the contribution from convexity must therefore always be caveated with rather lurid health warnings.

It is for this reason that we adopt in this study a quasi-model-agnostic approach. By this we mean that we do place ourselves in an affine-modelling framework, but we make our results as independent as possible from the specifics of any affine model (such as the number of factors, or the nature of the state variables—latent, specified, yield-curve-based,

^{*}Corresponding author. Email: riccardo.rebonato@edhec.edu

macroeconomic, etc). We only require that *some* affine model should exist, capable of recovering the market yield curve and yield covariance matrix with the precision required by our study. We observe that at least one such model must exist, and we then rely in our reasoning on a simple equivalence result that we state and briefly prove in section 2.

We put our quasi-model-agnostic methodology to the test by examining whether it is possible to enter a trading strategy with an attractive Sharpe Ratio based on

- (1) immunizing against the exposure to ‘level’ and ‘slope’ risk;
- (2) estimating the yield volatilities over the next time interval;
- (3) comparing this estimate with the volatility ‘implied’ by the curvature in the yield curve; and
- (4) going ‘long’ or ‘short’ convexity as our model suggests.

As for point 2, on purpose we do not employ a particularly sophisticated method to estimate the next-step volatility, because we want to ascertain whether easily available information about future volatility is correctly embedded in the shape of the yield curve.

As we show in section 4, the convexity-induced downward curvature of the yield curve is analogous to an ‘option premium’ (in the form of a ‘yield give-up’) that an investor has to pay to enjoy the benefits of convexity. Exactly as in option trading, the long-convexity investor will be compensated for this ‘yield cost’ by making money during her re-hedging activity (‘buying low’ and ‘selling high’). If the realized volatility of yields matched on average the ‘implied volatility’ embedded in the shape of the yield curve, there would be no money to be made on average from being either long or short convexity. Is this the case?

By looking at 30 years of data, we find that neither the strategy of being systematically long or short convexity (and immunized against ‘level’ and ‘slope’ risk) would have been profitable. However, a *conditional* strategy, a strategy, that is, that looks at the difference between the ‘implied’ and the statistically estimated volatilities, would have identified extended periods during which the convexity was not correctly priced (by which we mean that the strategy of being conditionally long or short convexity, while remaining hedged against the yield curve factors, would have had an attractive Sharpe Ratio).

The paper is organized as follows.

In section 2 we state clearly the assumptions that underlie our work, and introduce the notation.

In section 3 we introduce the concept of equivalent affine models. This will be useful in what follows because, for our results to hold, we only require that *some* affine model should exist, capable of recovering with the desired precision the market yield curve and the yield covariance matrix. We show that one such models exists, and therefore we can rest assured that any equivalent affine model will produce exactly the same results.

Before delving in the theoretical analysis, we ask in section 4 whether there is empirical evidence that the volatility of yields affects their prices. We answer in the affirmative, and we give an economic interpretation to the link we discover between the yield volatilities and the curvature of the yield curve.

Now the theoretical treatment begins. It is carried out with the two levels of assumptions stated in section 2. More precisely, in section 5 we derive a first expression for convexity in an affine setting under the assumption of constant-maturity yields. We only use this assumption in sections 5 and 6 to simplify the treatment and enhance intuition.

In section 6 we provide a simple (albeit approximate) graphical construction (a table of trade-offs between ‘yield give-up’ and ‘convexity pick-up’) that allows the reader to gain a clear qualitative understanding of the nature of the trade-offs in a convexity trade.

In section 7 onwards we relinquish the constant-maturity assumption, and we start our general affine treatment. The expression we obtain for convexity in this setting is still specific-model-dependent.

In sections 8 and 9 we then embark on the tasks (i) of relinquishing the dependence of the value of convexity on the features of a specific affine model, and (ii) of linking it to observable features of the yield curve shape, and on measurable yield volatilities. This is done in two different ways (sections 8.1 and 8.2), which offer different financial interpretations in terms of yield volatilities and of well-known market quantities such as ‘carry’ and ‘roll-down’ (which are now given a precise definition).

In section 10 we show how the remaining vestigial remnants of a specific affine model (the delta-neutralizing weights) can also be made to disappear, thereby making the approach truly specific-affine-model independent.

We can now test empirically our findings. We describe the data and the empirical methodology in section 11, and present and critically discuss the empirical results in section 12.

Section 13 concludes the paper.

2. Assumptions and notation

In order to derive our results we must place ourselves in an affine setting, i.e. we must assume that, for *some* state variables, it should be possible to express yields (and the short rate[†]) as an affine function of the state variable themselves. This assumption is needed in section 8, where we link the ‘value of convexity’ to observable features of the yield curve and to yield volatilities.

We stress that, as shown in section 3, it is not necessary to specify the precise nature of the state variables, or their precise dynamics. One must only assume that some affine model exists capable of recovering to the desired degree of precision the shape of the yield curve and the yield volatilities. We know that one such model exists because, by using principal components as state variables—as commonly done in the modern affine literature[‡]—we can approximate an exogenous yield curve and covariance matrix to an arbitrary degree of accuracy.

In our general treatment we do *not* work with constant-maturity yields. We only do so in sections 5 and 6 to help the reader’s intuition, and keep the derivations simple. From

[†]We require the short rate to be an affine function of the mean-reverting state variables because we want the discount function to be an exponential affine function of the state variables.

[‡]see, e.g. Adrian *et al.* (2014) or Joslin *et al.* (2014).

section 7 onwards the constant-maturity-yield assumption is no longer enforced.

We denote the n unspecified state variables by $\{x\}$, and we impose that they should follow a mean-reverting dynamics of the type

$$d\vec{x}_t = \mathcal{K}(\vec{\theta} - \vec{x}_t)dt + Sd\vec{z}_t \quad (1)$$

where $\mathcal{K}$ is the $n \times n$ reversion speed matrix, $\vec{\theta}$ is an $n \times 1$ vector, S an $n \times n$ diffusion matrix and $d\vec{z}_t$ an $n \times 1$ vector of Brownian shocks with

$$\mathbb{E}[dz_t^i dz_t^j] = \delta_{ij}dt, \quad (2)$$

and with δ_{ij} denoting the Kronecker delta. An expression such as x' denotes the transpose of x (where x can be a matrix or a vector).

We need the yield curve (i.e. the continuum of discount bond prices P_t^T) to be a continuous differential function of maturity, T , which it will automatically be given the affine assumption above.

For the derivations starting in section 8, we require time homogeneity, i.e. no explicit dependence on calendar time alone. This is not unduly restrictive, because convexity is only economically significant for very long maturities (say, over ten years) and over these horizons expectations (which can in general introduce a calendar-time dependence) matter very little.

Finally, for our general results we require the concept of equivalent affine model, which we introduce in the next section.

We stress that in section 6—and only in section 6—we introduce an additional hypothesis (not needed for the general setting), namely that yield volatilities should be perfectly correlated. This assumption is only introduced in order to derive a simple indicator of the ‘value of convexity’, namely the break-even volatility, which in turn gives a very transparent graphical illustration of the trade-off between convexity and ‘yield give-up’ shown in figures 4 and 5.

3. Equivalent affine models

In this section we show that, under mild conditions, all affine models that produce exactly the same yield curve and the same covariance matrix display an exactly identical dynamic behaviour—and, in particular, produce exactly the same predictions for convexity.

More precisely, let $\{m_1\}$ and $\{m_2\}$ be two time-homogeneous, non-stochastic-volatility, essentially affine models with an invertible mapping from state variables to yields. Their state variables, $\{x_1\}$ and $\{x_2\}$, can be very different in the two models, and either, both or neither could include macro variables. The two models will be said to be ‘equivalent’ if at time t_0 (‘today’) they produce *exactly* the same yield curve and the same yield covariance matrix.[†]

[†]If the two models have to reproduce *exactly* the same yield curve and the same yield covariance matrix, then, barring degeneracies, they cannot have a different number of state variables, nor can these be essentially different. We will extend the reasoning to quasi-equivalent models, however, for which different types and numbers of state variables can produce vanishingly small differences in the initial yield curve and in the model covariance matrix.

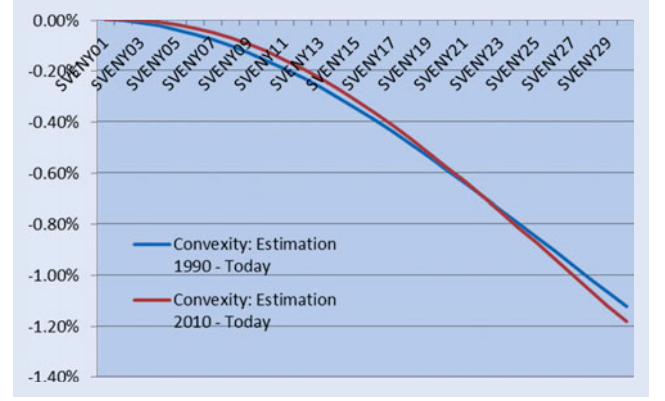


Figure 1. The value of convexity, i.e. the difference between the market yields and the yields that would prevail in the absence of convexity, as estimated in August 2015 using a three-factor principal-component-based model such as the one by [Rebonato et al. \(2014\)](#), for yields ranging from 1 year (point labelled ‘SVEN01’), to 29 years (point labelled ‘SVEN29’). The estimates of convexity from this model are reliable because the model accurately recovers (by construction) the volatilities of all yields. The two curves show that the ‘value of convexity’ is modestly affected by the precise choice of the historical period during which the yield volatilities are calculated, and to which the model is calibrated. As we discuss in section 2, any principal-component-based model would have produced essentially the same results.

Since the two equivalent models are affine, and we have assumed invertibility, it is always possible to find matrices A_1 and A_2 such that the yields can be expressed as functions of the state variables:

$$y_t = A_1 x_1^t \quad (3)$$

and

$$y_t = A_2 x_2^t. \quad (4)$$

This means that, if the $\{m_i\}$ model has $n^{(i)}$ factors, we can use the $n^{(i)}$ yields from the matrix inversion as state variables. (See [Ludvigson and Ng \(2009\)](#) for a discussion of the implications of this choice.)

Consider now the Partial Differential Equation (PDE) that governs the evolution of the yields in the $\mathbb{Q}$ (pricing, risk-neutral) measure. We don’t need to know the form of this PDE, but only that its solution is unique given the same initial conditions, the same drift and the same covariance structure.

But, given two equivalent models,

- the initial conditions are the same (because the two models produce the same time- t_0 yield curve by definition of equivalent models);
- the drift of yields or forward rates under $\mathbb{Q}$ is the same, because—this is the essence of the [Heath et al. \(1989\)](#) result—their no-arbitrage drifts are purely a function of the forward-rate covariance matrix (which is identical for the two models, by the equivalence assumption today, and in the future by time-homogeneity);
- the stochastic evolution is the same also because of the assumed equivalence (same covariance matrix) and time-homogeneity.

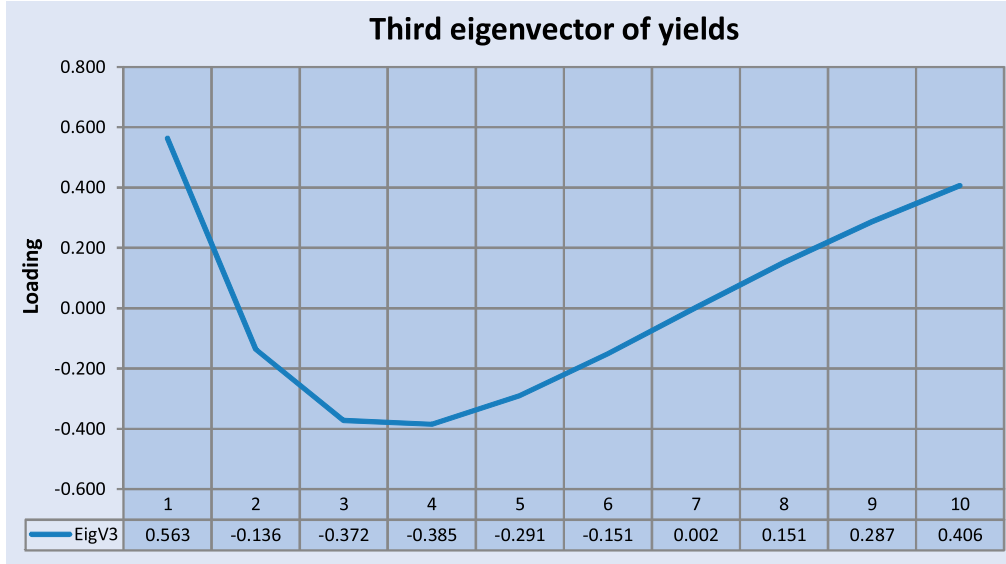


Figure 2. The loadings on the first 10 yields that make up the third eigenvector from the orthogonalization of the yield covariance matrix. Data from Gurkaynak *et al.* (2007), spanning the period 16-Nov-1971 to 1-Oct-2015.

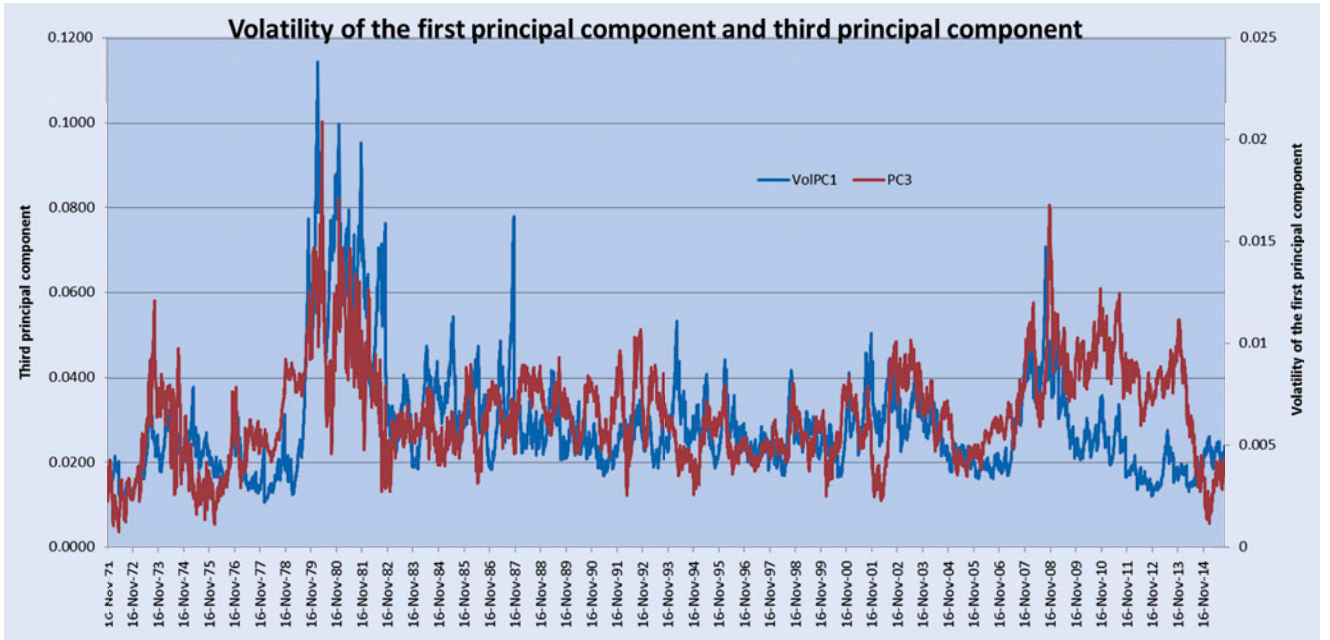


Figure 3. The time series of the volatility of the first principal component (right-hand vertical axis) and of the third principal component (left-hand vertical axis). Note the particularly strong correlation between the two curves during periods of high volatility.

From this it follows that the evolution of the yield curve under $\mathbb{Q}$ is, in absolutely all its aspects, exactly the same for any two equivalent models.

It is possible to show that, under the assumptions above, the two equivalent essentially affine models would behave identically also under the objective ($\mathbb{P}$) measure, but we do not need this result here.

As long as the mild conditions above about invertibility and non-stochastic volatility are satisfied, all conclusions we draw about one equivalent model therefore extend to any other equivalent model.

We next define a reference family of affine models to be *quasi-equivalent* to a given model if the differences in the initial yield curve and covariance matrix (defined over a possibly

very large, but finite, number of yield points) can be made vanishingly small by increasing the number of state variables in the models of the reference family. Take, for instance, the family of affine models that use principal components as state variables. Then, for any well-behaved initial yield curve and covariance matrix, it is always possible to approximate both these quantities as closely as desired by adding higher and higher principal components to the set of state variables. Therefore the family of principal-component-based affine models is quasi-equivalent to an arbitrary affine model with well-behaved initial yield curve and covariance matrix.

This is interesting, because principal-components-based affine models (such as the models by Diebold and Rudebush (2013), Adrian *et al.* (2014), Joslin *et al.* (2011), or by Rebonato



Figure 4. The ‘yield give-up’ (shown by the vertical downward-pointing arrow) from the stylized strategy discussed in the text for the case of a positive-convexity strategy, and with a typical (concave) shape of the yield curve. The funding advantage is a convex combination of the funding advantages of the different constituent bonds. The holdings in the bonds of different maturities cannot all have the same sign if one wants to immunize against the factor uncertainty. Therefore, the portfolio funding advantage (its yield give-up) can, in general, have either sign. For the case depicted in the figure and the stylized example discussed in the text, it must be negative.

et al. (2014)) can be made to recover both the market yield curve and the covariance matrix with arbitrary precision (and greater ease than most other models). As a consequence, at least one model capable of recovering the initial yield curve and covariance matrix (and hence yield volatilities) to an arbitrary degree of accuracy exists. But we show in section 3 (and more precisely in section 5) that in an affine setting the ‘value of convexity’ only depends on the correct recovery of the yields and of their volatilities.[†] Therefore the value of convexity produced by any one of the models that can recover the initial yield curve and covariance matrix to arbitrary precision becomes both model-independent and ‘correct’.[‡]

In this study we have therefore obtained our convexity results with a principal-component-based model (for instance, such as the one by *Rebonato et al.* (2014)) in the back of our mind. However, our results will not depend on the specific parametrization of this ‘class-representative’ model or on this particular choice of state variables, as long as the statistically determined covariance matrix and the market yields are accurately recovered. Indeed, we show that the ‘correct’ (in the sense above) value of convexity can be determined without explicit reference to the parameters of any affine model. It is in this sense that our approach is quasi-model-agnostic.

Before delving into the modelling part of the paper, it is important to ask a very fundamental question. Figure 1 shows the *theoretical* value of convexity. But is there *prima facie* evidence that volatility does indeed affect the prices of Treasury bonds? Absent this relationship, the whole concept of the ‘efficient pricing’ of convexity would have to be revisited. We therefore look at this ‘foundational’ aspect in the next section.

[†]If we use a principal-component-based affine model with n factors, we are only guaranteed that n yields and their covariance elements will be exactly recovered. In practice, with a judicious choice of model parameters, it is possible to approximate the whole yield curve and the whole covariance matrix out to 30 years extremely closely. *Rebonato et al.* (2014) show in detail how this can be done.

[‡]It is only ‘correct’, of course, in a time-homogeneous, non-stochastic-volatility universe—hence the scare quotes around the adjective ‘correct’.

Yield Give-Up > 0		Yield Give-Up < 0	
$C > 0$	Enter strategy if $\sigma^2 > \frac{y}{C} > 0$	$\sigma^2 > \frac{y}{C} < 0 \Rightarrow$	always enter the strategy
$C < 0$	$\sigma^2 < \frac{y}{C} < 0$ (impossible)	Enter reverse strategy if $\sigma^2 < \frac{y}{C} > 0$	$\Rightarrow$ never enter the strategy

Figure 5. The optimal strategy in the four possible configurations corresponding to convexity pick-up positive or negative, and yield give-up positive or negative.

4. The link between volatility and the curvature of the yield curve

If convexity is ‘priced’, its effect should be to lower the yield of long-dated bonds. As we show in the following, the magnitude of the effect should depend (quadratically) on the yield curve volatility. As it is well known, volatility is highly variable. Are periods of greater volatility associated with more curved term structures of yields? Is there indeed a detectable relationship between the curvature of the yield curve and its volatility? These are the questions we address in this section.

Since both the curvature and the volatility of the yield curve are somewhat vague concepts, in order to make these questions operationally precise we proceed as follows.

First, given the well-known interpretation of the first three principal components as level, slope and curvature of the yield curve, we use the third eigenvector obtained from the orthogonalization of the yield covariance matrix[§] as a simple proxy for the yield curve curvature. Figure 2 shows the reasonableness of this choice.

As a next step, we estimate the time series of the first three principal components, obtained by orthogonalizing the covariance matrix of yield *levels*. Next, from the time series thus obtained, we estimate (using a simple GARCH(1,1) model) their absolute (not percentage) volatilities, $\sigma_{PC(i)}^t$, $i = 1, 2, 3$. Since this is not a predictive, but a descriptive, regression, we use the whole sample to estimate these volatilities.

Then we regress, one by one, the three volatilities thus obtained, $\sigma_{PC(i)}^t$, $i = 1, 2, 3$, against the first three principal components:

$$\sigma_{PC(i)}^t = \alpha_{(i)} + \vec{\beta}_{(i)}' \vec{PC}_t + \epsilon_{t+1}^{(i)}. \quad (5)$$

To be clear: ultimately, we would like to establish whether the results of the regressions are compatible with a causal link from volatility of the yield to the curvature of the yield curve. Given this, it may seem more natural to regress the curvature against the volatilities. However the volatilities of the first three principal components are strongly correlated (the average off-diagonal correlation among the three volatilities is 77.6%, and the highest 86.5%), and we therefore prefer to use volatilities as left-hand variables, and the orthogonal principal components as regressors. This orthogonality will then allow a very clean decomposition of the R^2 from the multivariate regression into

[§]The first three eigenvectors are almost identical irrespective of whether one uses yield levels or yield changes. We used the data in *Gurkaynak et al.* (2007) to obtain the covariance matrix among the zero-coupon yields.

Table 1. The intercepts ($\alpha_{(i)}$, second row) and slopes ($\beta_{(i)}^j$, third to fifth row) of the left-hand variables displayed in the first row of the second, third and fourth column in the multi-variate regression (5). The numbers in parentheses next to the regression coefficients denote the standard error. The R^2 for each regression is reported in the first row. Note the much larger magnitude of the beta coefficients on the third principal component (last row, in bold).

	$\sigma_{PC(1)} (R^2 = 0.455)$	$\sigma_{PC(2)} (R^2 = 0.375)$	$\sigma_{PC(3)} (R^2 = 0.389)$	
$\alpha_{(i)}$	-0.002 (0.000)	0.000 (0.000)	-0.001 (0.000)	
$\beta_{(i)}^{PC1}$	0.065 (0.001)	0.015 (0.000)	0.013 (0.000)	
$\beta_{(i)}^{PC2}$	0.069 (0.090)	0.006 (0.003)	-0.004 (0.002)	
$\beta_{(i)}^{PC3}$	2.575 (0.039)	0.901 (0.013)	0.435 (0.000)	(6)

Table 2. The results from running the nine univariate regressions of the three volatilities against the three principal components. The three columns refer to the first three principal components as regressors, and the left-hand variables are the volatilities of the principal components.

σ_1	$\begin{bmatrix} PC_1 \text{ as regressor} \\ a = 0.02 (0.00) \\ b = 0.07 (0.01) \\ R^2 = 0.23 \end{bmatrix}$	σ_2	$\begin{bmatrix} PC_1 \text{ as regressor} \\ a = 0.01 (0.00) \\ b = 0.02 (0.00) \\ R^2 = 0.12 \end{bmatrix}$	σ_3	$\begin{bmatrix} PC_1 \text{ as regressor} \\ a = 0.00 (0.00) \\ b = 0.01 (0.00) \\ R^2 = 0.24 (0.00) \end{bmatrix}$	
σ_1	$\begin{bmatrix} PC_2 \text{ as regressor} \\ a = 0.03 (0.00) \\ b = 0.07 (0.01) \\ R^2 = 0.00 \end{bmatrix}$	σ_2	$\begin{bmatrix} PC_2 \text{ as regressor} \\ a = 0.01 (0.00) \\ b = 0.01 (0.00) \\ R^2 = 0.00 \end{bmatrix}$	σ_3	$\begin{bmatrix} PC_2 \text{ as regressor} \\ a = 0.00 (0.00) \\ b = -0.00 (0.00) \\ R^2 = 0.00 \end{bmatrix}$	(7)
σ_1	$\begin{bmatrix} PC_3 \text{ as regressor} \\ a = 0.01 (0.00) \\ b = \mathbf{2.57} (0.05) \\ R^2 = 0.22 \end{bmatrix}$	σ_2	$\begin{bmatrix} PC_3 \text{ as regressor} \\ a = 0.03 (0.00) \\ b = \mathbf{0.90} (0.02) \\ R^2 = 0.25 \end{bmatrix}$	σ_3	$\begin{bmatrix} PC_3 \text{ as regressor} \\ a = 0.00 (0.00) \\ b = \mathbf{0.44} (0.01) \\ R^2 = 0.15 \end{bmatrix}$	

its individual marginal components. Since the choice of left- and right-hand variables does not imply any causality, the two approaches are conceptually equivalent.

From the work by [Duguillaume et al. \(2013\)](#) we know that there is a (rather complex) relationship between volatilities and the level of rates. This, however, is not what we are interested in. We focus our attention on whether there is a relationship between the level of volatility of the curvature of the yield curve.

As tables 1 and 2 show, we find that this is indeed the case, and that, for each of the volatilities, the link is strongest with the curvature:

When we run the nine univariate regressions of the three volatilities against the three principal components we find that there is no link between the slope and the volatilities; that, as well known, there is a relationship between the absolute volatility (especially of the first principal component) and the level; finally—and this is the finding of relevance for this study—that *there is a clear dependence between the volatilities of all the principal components and the curvature of the yield curve.*

In particular, we find that

- the volatility of the level is linked to the level and the curvature;
- the volatility of the slope is linked to the curvature;
- the volatility of the curvature is linked to the level and the curvature.

So, the volatilities of all the principal components are linked to the curvature. The curvature is the only principal component with a significant loading on all the principal components.

Figure 3 shows the time series of the volatility of the first principal component (which represents the lion's share of the volatility of the yield curve) and the time series of the third principal component. At a qualitative level one clearly notices the particularly strong correlation between the two time series during periods of high volatility.

In sum: we have found that the volatility of every mode of deformation of the yield curve is related to the curvature. At a phenomenological level we have therefore established that a link does exist between the curvature and the volatility of the yield curve. It remains to be seen whether this relationship behaves as term-structure theory suggests it should. We turn to this question in the rest of the paper.

5. The expression for convexity in an affine setting

In this section we will enforce for the sake of notational and conceptual simplicity the constant-maturity-yield approximation, an approximation that will be removed in the treatment that starts in section 7. Within the limits of this approximation we provide a framework to gain an intuitive financial interpretation of the ‘value of convexity’—an interpretation that we pursue in section 6.

We place ourselves in an affine setting. More precisely, consider a set of $n - 1$ state variables, $\vec{x}_t$, following a joint

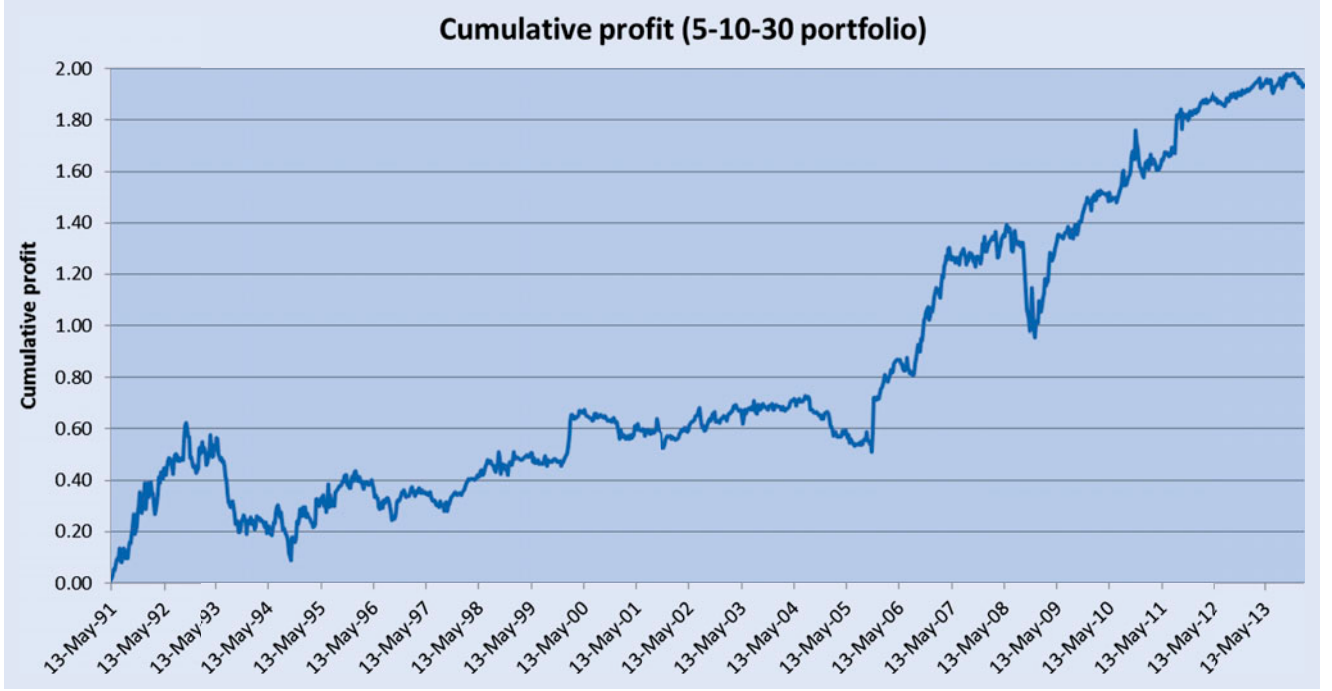


Figure 6. The cumulative profit from the convexity strategy with Portfolio 5/10/30 for the period August 1991 to January 2015. The cumulative profit or loss (P&L) shown on the y axis refer to being long or short in the longest-maturity (30-year) bond exactly one unit of currency.

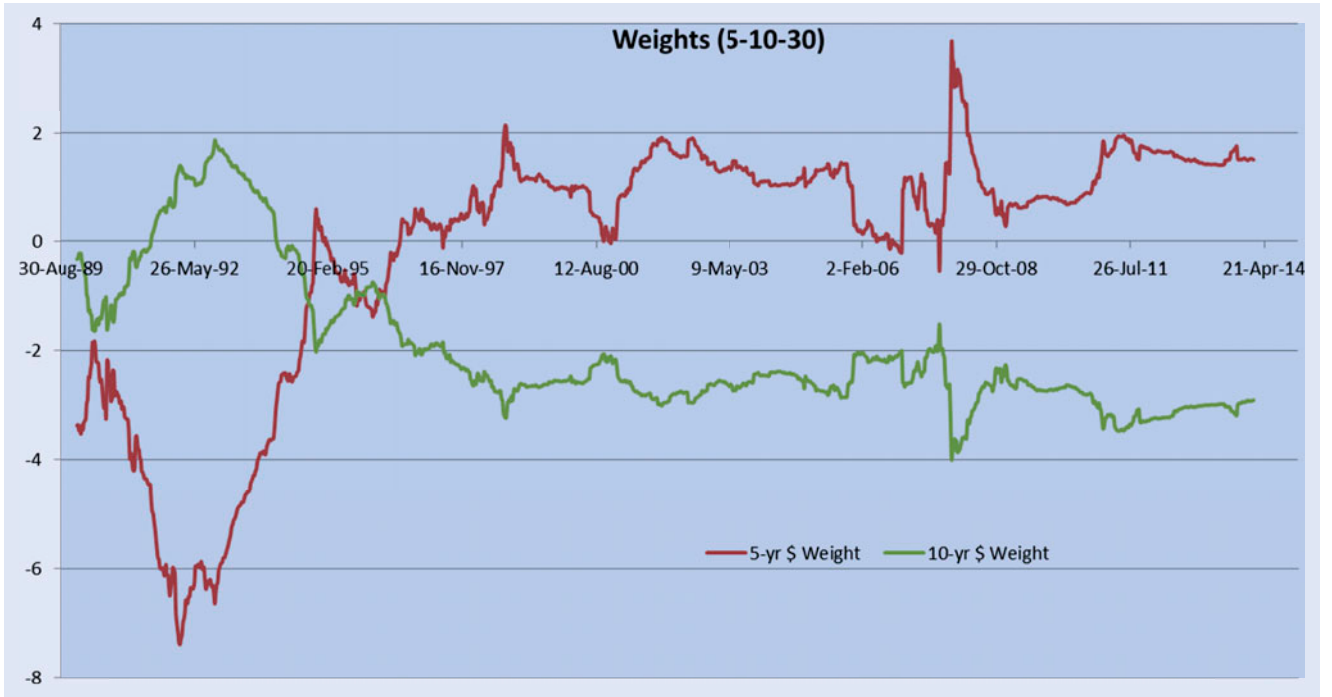


Figure 7. The weights of the 5-year and of the 10-year bonds throughout the strategy. The weight of the 30-year bond was always taken to be 1.

mean-reverting processes of the form

$$d\vec{x}_t = \mathcal{K}(\vec{\theta} - \vec{x}_t)dt + Sd\vec{z}_t \quad (8)$$

with

$$\mathbb{E}[dz_t^i dz_t^j] = \delta_{ij}dt, \quad (9)$$

with δ_{ij} denoting the Kroeneker delta. In what follows we will assume these state variables to be principal components of

yields. Let these state variables be related to the short rate by an affine transformation of the form

$$r_t = u_r + \vec{g}'\vec{x}_t. \quad (10)$$

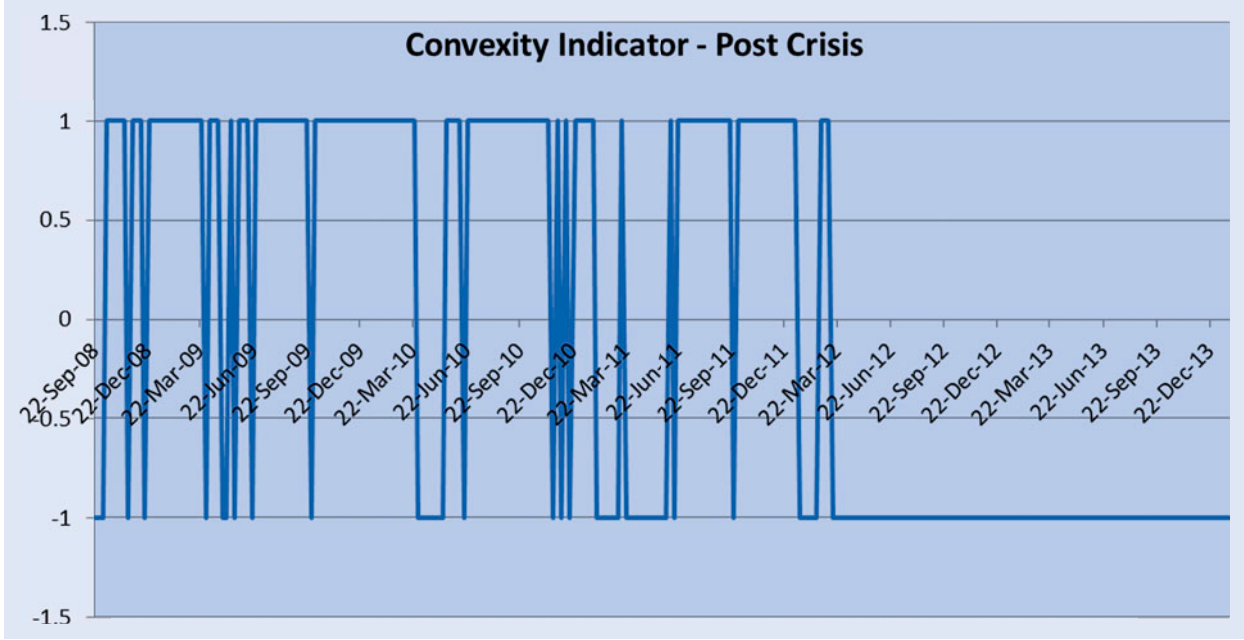


Figure 8. The convexity indicator for the post-crisis period.

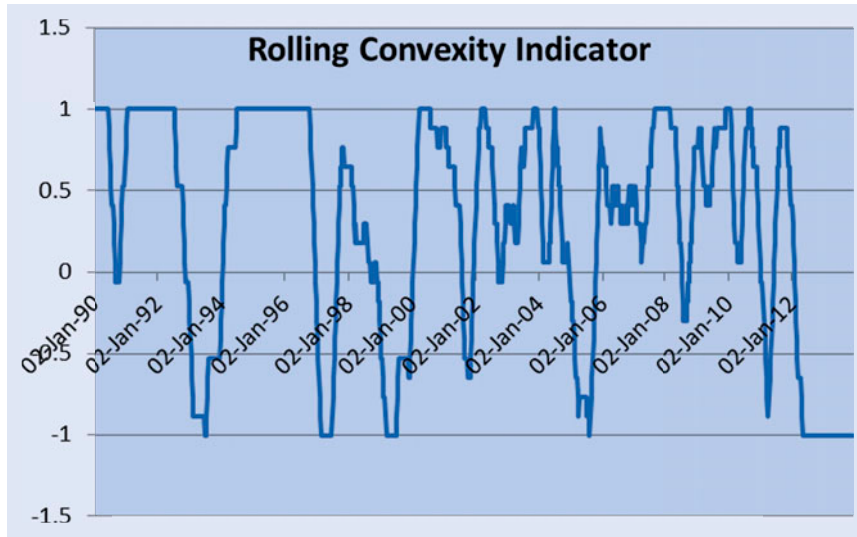


Figure 9. The three-week rolling average of the convexity indicator for the whole period.

Then it is well known (see, e.g. Dai and Singleton 2000) that the bond prices, P_t^T , will be given by an expression of the type

$$P_t^T = e^{A_t^T + (\vec{B}_t^T)' \vec{x}_t}. \quad (11)$$

Our strategy will be to construct portfolios made up of n bonds immunized ('duration-hedged') against the sources of risk in the economy (recall that $n - 1$ is the number of factors in the process (8)).

Given the expression above for the bond price, remembering the definition

$$y_t^T = -\frac{1}{T-t} \log P_t^T = -\frac{1}{T-t} \left(A_t^T + (\vec{B}_t^T)' \vec{x}_t \right) \quad (12)$$

and denoting by y_k the T_k -maturity yield, $y_k \equiv y_t^{T_k}$, we easily obtain for the k th yield[†]

$$dy_k = -\frac{1}{T_k - t} \left(\vec{B}_t^{T_k} \right)' d\vec{x}_t \equiv \left(\vec{b}_t^{T_k} \right)' d\vec{x}_t. \quad (13)$$

From this it follows that

$$dy_k = \left(\vec{b}_t^{T_k} \right)' \left[\mathcal{K} \left(\vec{\theta} - \vec{x}_t \right) dt + S d\vec{z}_t \right] \quad (14)$$

and therefore the $n - 1$ components of the volatility of the k th yield are given by the elements of the vector $(\vec{\sigma}^{y_k})^T$:

$$(\vec{\sigma}^{y_k})' = \left(\vec{b}_t^{T_k} \right)' S. \quad (15)$$

[†]This is the differential of a constant-maturity yield. We derive expressions including the time decay in section 6.

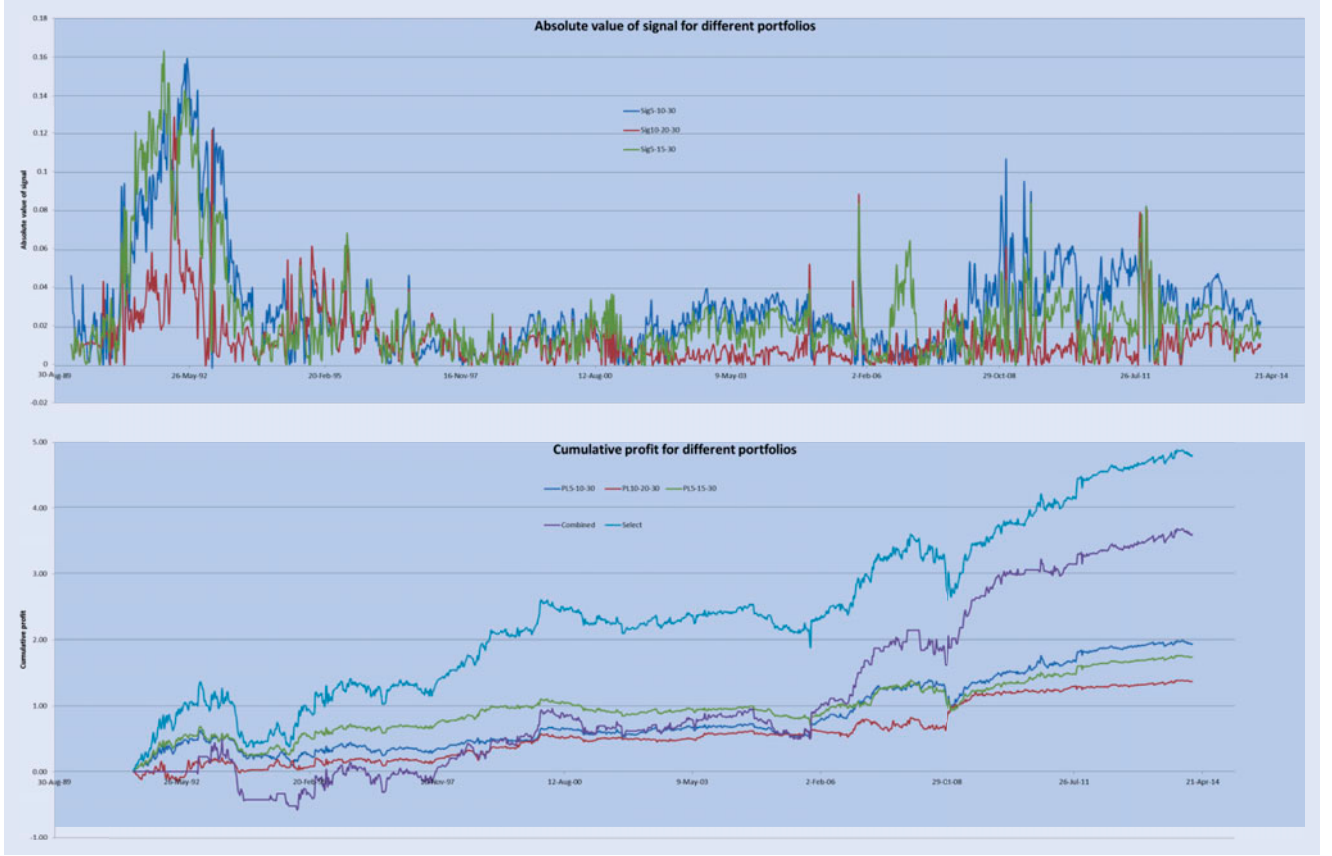


Figure 10. The strength of the signal (upper panel) against the cumulative P&L for the various strategies. The construction of the curves labelled ‘Combined’ and ‘Select’ is explained in the text.

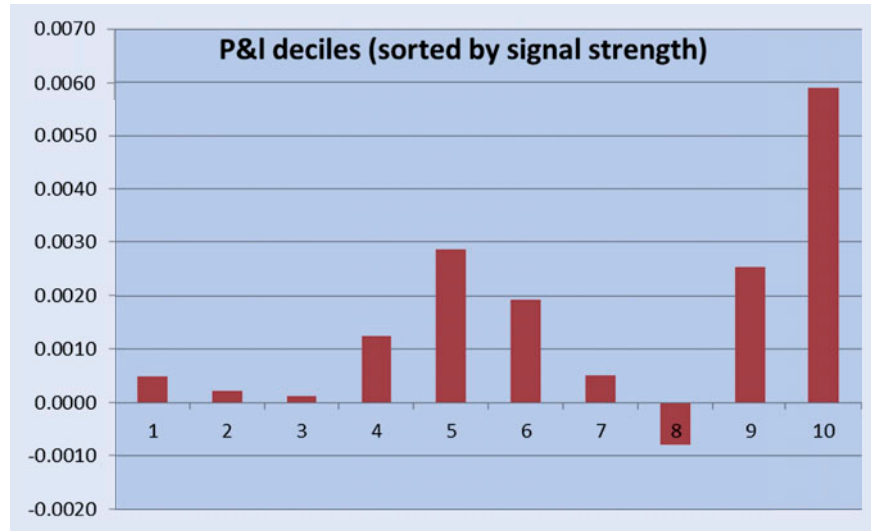


Figure 11. Each bar represents the average P&I in the corresponding decile. The deciles were obtained by sorting the (signed) realized P&L by (absolute) signal strength.

The total volatility of the k th yield, which we denote by σ_k , is given by

$$\sigma_k^2 = (\vec{\sigma}^{y_k})' (\vec{\sigma}^{y_k}) . \quad (16)$$

As it is well known, by constructing a portfolio of n bonds, making it riskless, and imposing no arbitrage one obtains the equation

$$\sum \omega_i \left[\left(\frac{\partial P_t^{T_i}}{\partial t} + \frac{\partial P_t^{T_i}}{\partial y_t^{T_i}} \mu_{y_t^{T_i}} - P_t^{T_i} r_t \right) + \underbrace{\frac{1}{2} \frac{\partial^2 P_t^{T_i}}{\partial (y_t^{T_i})^2} \sigma_i^2}_{\text{convexity}} \right] = 0, \quad (17)$$

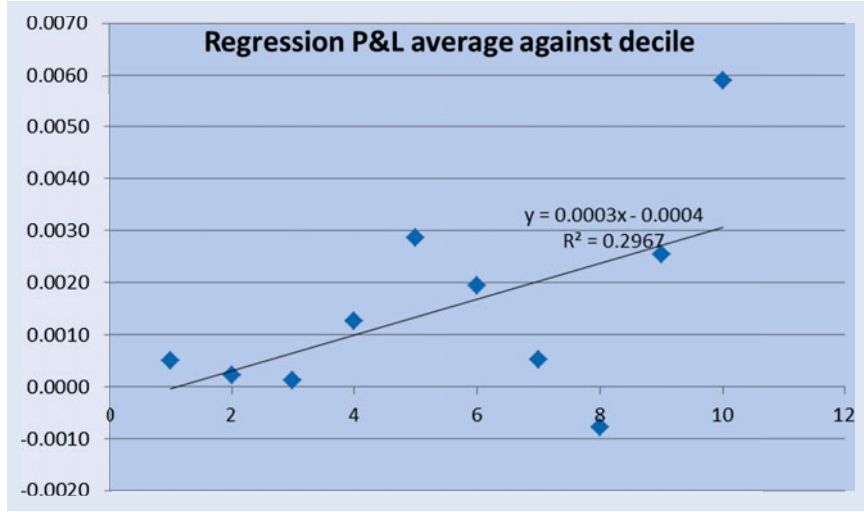


Figure 12. The average P&L in each decile plotted against the decile. The figure also shows the trend line and the R^2 statistic.

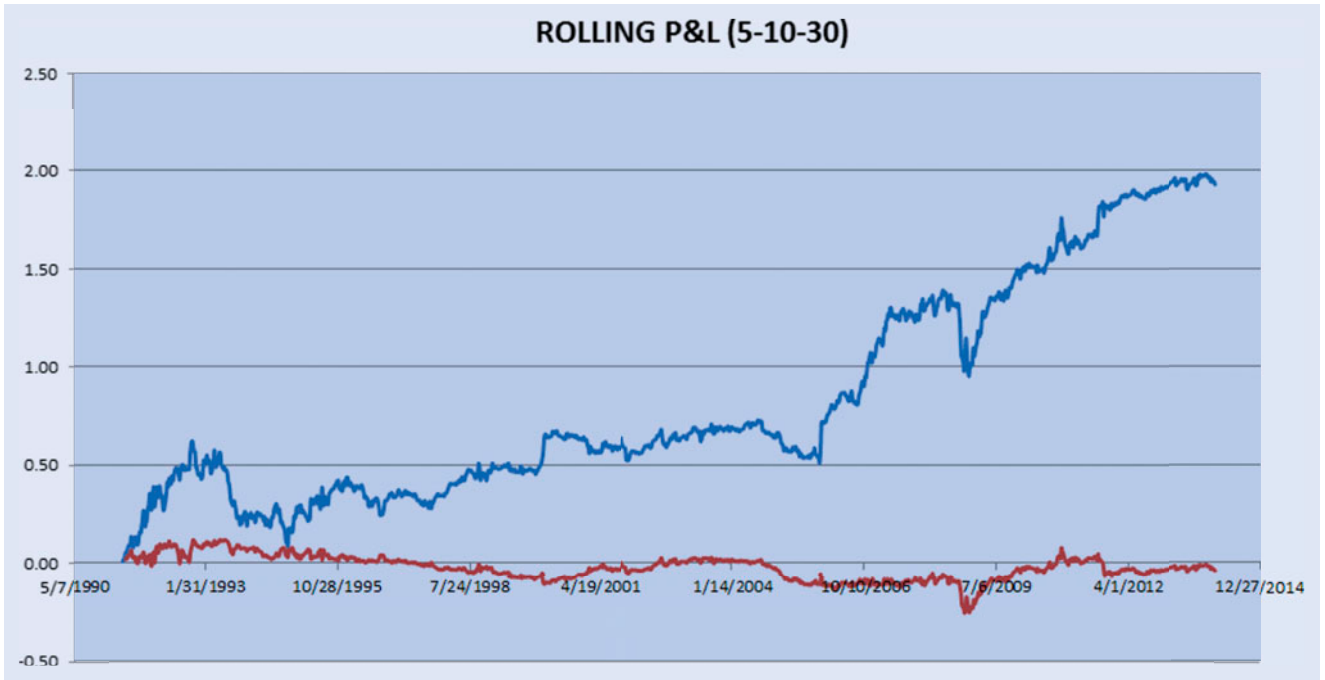


Figure 13. The cumulative profit from the 5-10-30 strategy and the cumulative P&L explained by the residual duration exposure.

where ω_i are the optimal (duration-neutralizing) weights and $\mu_{y_t^{T_i}}$ is the time- t drift of the T_i -maturity yield.

If we denote the vector of weights by $\vec{\omega}$, $\vec{\omega} = [\omega_1 \ \omega_2 \ \dots \ \omega_n]'$, it is straightforward to show that equation (17) can be written more compactly as

$$\vec{\omega}' (\vec{c}_1 + \vec{c}_2 + \vec{c}_3 + \vec{c}_4) = 0 \quad (18)$$

with

$$\vec{c}_1 = \begin{bmatrix} \frac{\partial P_t^{T_1}}{\partial t} \\ \frac{\partial P_t^{T_2}}{\partial t} \\ \dots \\ \frac{\partial P_t^{T_n}}{\partial t} \end{bmatrix} = \begin{bmatrix} P_t^{T_1} y_1 \\ P_t^{T_2} y_2 \\ \dots \\ P_t^{T_n} y_n \end{bmatrix}; \quad (19)$$

$$\begin{aligned} \vec{c}_2 &= \begin{bmatrix} \frac{\partial P_t^{T_1}}{\partial y_t^{T_1}} \mu_{y_t^{T_1}} \\ \frac{\partial P_t^{T_2}}{\partial y_t^{T_2}} \mu_{y_t^{T_2}} \\ \dots \\ \frac{\partial P_t^{T_n}}{\partial y_t^{T_n}} \mu_{y_t^{T_n}} \end{bmatrix} \\ &= \begin{bmatrix} -T_1 P_t^{T_1} (b_t^{T_1})' \left[\kappa (\vec{\theta} - \vec{x}_t) dt \right] \\ -T_2 P_t^{T_2} (b_t^{T_2})' \left[\kappa (\vec{\theta} - \vec{x}_t) dt \right] \\ \dots \\ -T_n P_t^{T_n} (b_t^{T_n})' \left[\kappa (\vec{\theta} - \vec{x}_t) dt \right] \end{bmatrix}; \quad (20) \end{aligned}$$

$$\vec{c}_3 = \begin{bmatrix} -P_t^{T_1} r_t \\ -P_t^{T_2} r_t \\ \dots \\ -P_t^{T_n} r_t \end{bmatrix}; \quad (21)$$

and finally

$$\vec{c}_4 = \begin{bmatrix} \frac{1}{2} \frac{\partial^2 P_t^{T_1}}{\partial y_t^2} (b_t^{T_1})' S S' (b_t^{T_1}) \\ \frac{1}{2} \frac{\partial^2 P_t^{T_2}}{\partial y_t^2} (b_t^{T_2})' S S' (b_t^{T_2}) \\ \dots \\ \frac{1}{2} \frac{\partial^2 P_t^{T_n}}{\partial y_t^2} (b_t^{T_n})' S S' (b_t^{T_n}) \end{bmatrix}. \quad (22)$$

This equation seems to show clearly the model dependence not only of the value of the convexity (given by the term $\vec{c}_4$), but of the overall profit from holding the neutralized portfolio over a time step dt . (This profitability comes from an interplay of the time decay, $\vec{c}_1$, the drift term, $\vec{c}_2$, the funding cost of the portfolio, $\vec{c}_3$, and the convexity itself, $\vec{c}_4$.) However, recalling the relationships $\sigma_k^2 = (\vec{\sigma}^{y_k})^T (\vec{\sigma}^{y_k})$ and $(\vec{\sigma}^{y_k})' = (b_t^{T_k})' S$, one can see that at least the convexity term c_4 can be written purely as a function of quantities observable from today's yield curve, and from the yield volatilities:

$$\vec{c}_4 = \begin{bmatrix} \frac{1}{2} T_1^2 P_t^{T_1} \sigma_1^2 \\ \frac{1}{2} T_2^2 P_t^{T_2} \sigma_2^2 \\ \dots \\ \frac{1}{2} T_n^2 P_t^{T_n} \sigma_n^2 \end{bmatrix}. \quad (23)$$

Therefore,

- (1) any two deterministic-volatility affine models that recover identically the yield curve and all yield volatilities imply exactly the same value for convexity;
- (2) if a deterministic-volatility affine model exactly recovers the market prices and the yield volatilities over the next time interval will price convexity exactly.

Our final goal is to obtain a model-independent expression for the other terms ($\vec{c}_1$, $\vec{c}_2$ and $\vec{c}_3$) as well. To do this we will relinquish in section 7 the constant-maturity assumption made in equation (13) for expositional convenience. Before doing so, however, we draw on the approximate results obtained so far to give a suggestive financial interpretation of the 'value of convexity'.

6. An intuitive aside—simplifying the analysis

In this section, we take a brief detour to give an intuitive interpretation to the expression for convexity obtained so far. We stress that the remainder of the treatment will not make use of the approximations introduced in this and in the previous section. For continuity, the reader can skip to section 7.

6.1. A one-factor approximate indicator of the value of convexity

We can write for the volatility of each individual yield

$$\sigma_k = h_k \bar{\sigma}^y \quad (24)$$

with[†]

$$\bar{\sigma}^y = \frac{1}{n} \sum_{k=1,n} \sigma_k. \quad (25)$$

With these equations, we still acknowledge that the volatilities of yields of different maturities are all different, but we express them as a fraction (h_k) of their average volatilities, $\bar{\sigma}^y$. So far, we have not introduced any assumptions. Our next step, however, will be to argue that what varies stochastically will be the average level of volatility (the term $\bar{\sigma}^y$), and not the fractions (h_k). We are making this assumption in order to reduce our analysis of the value of convexity to a single indicator.

With this approximation we can then rewrite term c_4 as

$$\vec{c}_4 = (\bar{\sigma}^y)^2 \begin{bmatrix} \frac{1}{2} T_1^2 P_t^{T_1} h_1^2 \\ \frac{1}{2} T_2^2 P_t^{T_2} h_2^2 \\ \dots \\ \frac{1}{2} T_n^2 P_t^{T_n} h_n^2 \end{bmatrix}, \quad (26)$$

where the quantities (h_k^2) remain unchanged, and all the variability in yield volatilities is taken up by the term $\bar{\sigma}^y$. This helps the analysis because we can rewrite equation (18) as

$$\vec{\omega}' (\vec{c}_1 + \vec{c}_2 + \vec{c}_3 + \vec{c}_4) = 0 \implies$$

$$\begin{aligned} \vec{\omega}' (\vec{c}_1 + \vec{c}_2 + \vec{c}_3) &= -\vec{\omega}' \vec{c}_4 \\ &= -(\bar{\sigma}^y)^2 \vec{\omega}' \begin{bmatrix} \frac{1}{2} T_1^2 P_t^{T_1} h_1^2 \\ \frac{1}{2} T_2^2 P_t^{T_2} h_2^2 \\ \dots \\ \frac{1}{2} T_n^2 P_t^{T_n} h_n^2 \end{bmatrix} \end{aligned} \quad (27)$$

We then have

$$\begin{aligned} (\bar{\sigma}^y)^2 &= \frac{\vec{\omega}' (\vec{c}_1 + \vec{c}_2 + \vec{c}_3)}{\vec{\omega}' \begin{bmatrix} \frac{1}{2} T_1^2 P_t^{T_1} h_1^2 \\ \frac{1}{2} T_2^2 P_t^{T_2} h_2^2 \\ \dots \\ \frac{1}{2} T_n^2 P_t^{T_n} h_n^2 \end{bmatrix}}. \end{aligned} \quad (28)$$

Since the numerator and the denominator are just numbers, we can write these two numbers as

$$Y \equiv \vec{\omega}^T (\vec{c}_1 + \vec{c}_2 + \vec{c}_3) \quad (29)$$

and

$$C \equiv \vec{\omega}^T \begin{bmatrix} -\frac{1}{2} T_1^2 P_t^{T_1} h_1^2 \\ -\frac{1}{2} T_2^2 P_t^{T_2} h_2^2 \\ \dots \\ -\frac{1}{2} T_n^2 P_t^{T_n} h_n^2 \end{bmatrix}. \quad (30)$$

This allows us to derive the neat expression

$$(\bar{\sigma}^y)^2 = \frac{Y}{C}, \quad (31)$$

which expresses the 'break-even volatility', $\bar{\sigma}^y$, as the ratio of two quantities, which we interpret below.

[†]We could take the average of volatilities or of variances. The best choice depends on whether we think that volatilities or variances move more closely in parallel.

To provide this interpretation, we combine the vectors c_1 and c_3 into a new vector, c_Y :

$$\begin{aligned}\vec{c}_Y &= \vec{c}_1 + \vec{c}_3 = \begin{bmatrix} -P_t^{T_1} y_1 \\ -P_t^{T_2} y_2 \\ \dots \\ -P_t^{T_n} y_n \end{bmatrix} + \begin{bmatrix} P_t^{T_1} r_t \\ P_t^{T_2} r_t \\ \dots \\ P_t^{T_n} r_t \end{bmatrix} \\ &= \begin{bmatrix} P_t^{T_1} (r_t - y_1) \\ P_t^{T_2} (r_t - y_2) \\ \dots \\ P_t^{T_n} (r_t - y_n) \end{bmatrix} \quad (32)\end{aligned}$$

Each term $(r_t - y_i)$ is the difference between the rate at which the investor funds the purchase of a discount bond, (the ‘short rate’, r_t) and the yield of the bond (y_i)—in market parlance this is often referred to as the ‘funding advantage’. Therefore the ‘funding advantage’, FA_Π , of the whole portfolio is given by

$$FA_\Pi = \vec{\omega}' (\vec{c}_1 + \vec{c}_3) = \vec{\omega}' \begin{bmatrix} P_t^{T_1} (y_1 - r_t) \\ P_t^{T_2} (y_2 - r_t) \\ \dots \\ P_t^{T_n} (y_n - r_t) \end{bmatrix}. \quad (33)$$

This quantity tells us whether our portfolio has what in trading parlance is called a positive or negative ‘carry’.

A glance at the term $\vec{c}_2$,

$$\vec{c}_2 = \begin{bmatrix} -T_1 P_t^{T_1} (b_t^{T_1})' \left[\mathcal{K}(\vec{\theta} - \vec{x}_t) \right] \\ -T_2 P_t^{T_2} (b_t^{T_2})' \left[\mathcal{K}(\vec{\theta} - \vec{x}_t) \right] \\ \dots \\ -T_n P_t^{T_n} (b_t^{T_n})' \left[\mathcal{K}(\vec{\theta} - \vec{x}_t) \right] \end{bmatrix}, \quad (34)$$

then immediately shows this is a ‘drift term’. More precisely, it is a term that tells us in which direction the yields are headed over the next time interval *because of expectations*: the factors we have chosen as state variables for our affine model will deterministically revert towards their reversion levels (these are the terms $\left[\mathcal{K}(\vec{\theta} - \vec{x}_t) \right]$); these factor moves are then translated into yield moves via the ‘duration-related’ terms, $(b_t^{T_i})'$.

The profitability of the portfolio will depend on this drift term as well, but it is more difficult to interpret it intuitively: we will, of course, include it in our quantitative analysis, but we will ignore it when we give below a suggestive graphical interpretation of the profitability of a convexity strategy.

Going back to the break-even volatility, expression (31) provides a simple quantity, easy to monitor and easy to understand, that can tell us what the overall level of the yield volatilities should be over the next time interval for the convexity to be fairly priced. It also lends itself to a nice, if approximate, graphical interpretation, that we discuss below.

6.2. A graphical interpretation

As discussed, let us neglect for the moment the drift term $\vec{c}_2$. Then consider the following very stylized example: enter a naive long-convexity strategy, whereby the investor is long 50 units of the 30-year bond, long 50 units of the 10-year bond

and short 100 units of the 20-year bond. For simplicity, let’s unrealistically assume that the strategy is cash neutral (i.e. that the portfolio costs zero to set up).

For a typical shape of the yield curve, which tends to be concave at the long end, (and so with the 20-year yield higher than the average of the 10- and 30-year yields), the strategy will have a negative funding cost—by which we mean that the average yield of the bonds in the long position (the 10- and 30-year bonds) will be lower than the yield of the short bond (the 20-year bond). We call this funding cost the ‘yield give-up’.

At the same time, the portfolio will be long convexity (because convexity grows more than linearly with maturity, and therefore the average of the 10- and 30-year convexity will be higher than the convexity of the 20-year bond). The portfolio will display a ‘convexity pick-up’. The interplay between the two is shown in figure 4.

Is this convexity pick-up enough to compensate for the yield give-up? Within the simplifications we have made to arrive at this picture, equation (31) gives the answer, which will depend on comparing the prediction of the realized volatility over the next (re-hedging) interval with the ‘yield give-up’ term, as shown in figure 5.

The best configuration, of course, is found in the top right corner, where the investor can find a positive convexity pick-up and a positive funding advantage. The worst would be in bottom left corner, where the portfolio has negative carry *and* a negative convexity pick-up. Note however that, under the approximations made, this configuration cannot be attained because it would imply $\sigma^2 < 0$. The two entries on the main diagonal then suggest whether a long- or short-convexity strategy would be profitable.

We stress that the analysis presented here is suggestive rather than exact. In particular, the interpretation of the yield give-up given above is incomplete. We show in what follows how to make this intuition precise, and how to express the value of the whole strategy (i.e. all the terms $\vec{c}_1$, $\vec{c}_2$, $\vec{c}_3$ and $\vec{c}_4$, not just the convexity contribution) purely in terms of market observables.

7. A model-dependent expression for the profitability of convexity portfolios

We now return to the main flow of the study, and we no longer neglect any terms. In particular, we do not make the constant-maturity-yield approximation.

So far we have worked in term of yields in order to arrive at a simple intuitive expression for the ‘value of’ convexity and of the break-even volatility. To proceed, we now work directly with bond prices. Given the dynamics in equations (8)–(10), Ito’s lemma for the differential of the bond price gives:

$$dP_t^T = \frac{\partial P_t^T}{\partial t} dt + \left(\text{grad} P_t^T \right)' d\vec{x} + \frac{1}{2} \text{Tr} \left[S' B_t^T \left(B_t^T \right)' S \right] dt, \quad (35)$$

where the symbol $\left(\text{grad} P_t^T \right)'$ denotes the (transpose of the) gradient of the bond price with respect to the state variables:

$$\text{grad} P_t^T = \begin{bmatrix} \frac{\partial P_t^T}{\partial x_1} \\ \frac{\partial P_t^T}{\partial x_2} \\ \dots \\ \frac{\partial P_t^T}{\partial x_n} \end{bmatrix}. \quad (36)$$

As for the expression $\text{Tr}[S^T B_t^T (B_t^T)^T S]$, it denotes the trace of the matrix $[S^T B_t^T (B_t^T)^T S]$. The term $\frac{1}{2} \text{Tr}[S^T B_t^T (B_t^T)^T S] dt$ is the volatility-dependent convexity term.

Let $\{w_i\}$ denote the weights that ensure that the portfolio is first-order immunized, i.e. does not change value, to first order, when the factors move. As we saw, for immunization to hold the weights must satisfy the $n - 1$ equations

$$\sum_{i=1,n} w_i \frac{\partial P_t^{T_i}}{\partial x_k} = 0. \quad (37)$$

We can place each of the terms in equation (37) in a vector, $\vec{g}$, of $n - 1$ components:

$$g_k = \sum_{i=1,n} w_i \frac{\partial P_t^{T_i}}{\partial x_k}, \quad k = 1, 2, \dots, n - 1 \quad (38)$$

As usual, with these special weights the portfolio has become risk-less, and, under penalty of arbitrage, can therefore only grow at the risk-less rate of return, r_t . This gives the usual equation

$$\begin{aligned} & \sum_{i=1,n} w_i d P_t^{T_i} \\ &= \sum_{i=1,n} \left(w_i \frac{\partial P_t^{T_i}}{\partial t} dt + \frac{1}{2} w_i \text{Tr} \left[S' B_t^{T_i} (B_t^{T_i})' S \right] dt \right) \\ &+ (\vec{g})' d \vec{x} = r_t \left[\sum_{i=1,n} w_i P_t^{T_i} \right] dt. \end{aligned} \quad (39)$$

But because of equation (37), we have $(\vec{g})^T d \vec{x} = 0$, and therefore

$$\begin{aligned} & \sum_{i=1,n} w_i \left(\frac{\partial P_t^{T_i}}{\partial t} + \frac{1}{2} \text{Tr} \left[S' B_t^{T_i} (B_t^{T_i})' S \right] \right) \\ &= r_t \left[\sum_{i=1,n} w_i P_t^{T_i} \right]. \end{aligned} \quad (40)$$

Multiplying and dividing the left-hand side by $P_t^{T_i}$, we obtain

$$\begin{aligned} & \sum_{i=1,n} \omega_i \left(\frac{1}{P_t^{T_i}} \frac{\partial P_t^{T_i}}{\partial t} - r_t + \frac{1}{2} \frac{1}{P_t^{T_i}} \text{Tr} \left[S' (B_t^{T_i}) (B_t^{T_i})' S \right] \right) \\ &= 0 \end{aligned} \quad (41)$$

with the new weights, $\{\omega_i\}$, given by

$$\{w_i P_t^{T_i}\} = \{\omega_i\}. \quad (42)$$

If we now define the matrix, D , to be given by

$$D = \sum_{i=1,n} \omega_i (B_t^{T_i}) (B_t^{T_i})', \quad (43)$$

we have the following nice and compact expression for the overall convexity of the portfolio, $\mathcal{C}\mathfrak{P}$:

$$\mathcal{C}\mathfrak{P} = \frac{1}{2} \text{Tr}[S' D S] = \sum_{i=1,n} \omega_i \left(r_t - \frac{1}{P_t^{T_i}} \frac{\partial P_t^{T_i}}{\partial t} \right). \quad (44)$$

We note that the expression obtained so far is still model-dependent.

8. Theoretical portfolio convexity as a function of market observables

In this section we move to obtaining a model-independent expression for the theoretical value of the portfolio convexity. We present two equivalent expressions for this quantity.

8.1. The theoretical portfolio convexity as a function of forward rates

Recall that the instantaneous forward rate is given by

$$f_t^T = -\frac{\partial \log P_t^T}{\partial T} = \frac{\partial \log P_t^T}{\partial t} = \frac{1}{P_t^{T_i}} \frac{\partial P_t^{T_i}}{\partial t} \quad (45)$$

where the last line follows from time-homogeneity (i.e. from $-\frac{\partial}{\partial T} = \frac{\partial}{\partial t}$). We can therefore re-write the expression for $\mathcal{C}\mathfrak{P}$ obtained in the previous section (equation (44)), to obtain

$$\mathcal{C}\mathfrak{P} = \frac{1}{2} \text{Tr}[S' D S] = - \sum_{i=1,n} \omega_i (f_t^{T_i} - r_t). \quad (46)$$

This expression is very interesting because, apart from the weights, ω_i , the remaining terms on the right-hand side are completely model-independent and can be directly read from the shape of the yield curve today (assuming that a reasonable proxy for the ‘short rate’ can be observed). This establishes a first link between the shape of the yield curve and the theoretical portfolio convexity. We could stop here, but we want to establish a second link between convexity and observable yield-curve variables, because doing so enhances the interpretation of the ‘origin’ of convexity, and formalizes the intuitive concepts of ‘yield give-up’ and ‘convexity pick-up’ introduced in section 4.

8.2. The theoretical portfolio convexity as a function of ‘Carry’ and ‘Roll-Down’

We can express the right-hand side in equation (44) in terms of quantities (‘carry’ and ‘roll-down’) which form a prominent part of market lore, and which enhance the financial intuition above the value of convexity.

To this effect, consider again the term $\frac{1}{P_t^{T_i}} \frac{\partial P_t^{T_i}}{\partial t}$. From

$$P_t^T = e^{A_t^T + (\vec{B}_t^T)' \vec{x}_t} \quad (47)$$

we obtain

$$\begin{aligned} \frac{1}{P_t^T} \frac{\partial P_t^T}{\partial t} &= \frac{\partial \log P_t^T}{\partial t} = \frac{\partial}{\partial t} \left[A_t^T + \left(\vec{B}_t^T \right)' \vec{x}_t \right] \\ &= \frac{\partial A_t^T}{\partial t} + \left[\frac{\partial \left(\vec{B}_t^T \right)'}{\partial t} \vec{x}_t \right]. \end{aligned}$$

Since the yield of a bond is given by equation (12), differentiating both sides of $(T - t) y_t^T = - \left(A_t^T + \left(\vec{B}_t^T \right)' \vec{x}_t \right)$ with respect to t we get

$$y_t^T - (T - t) \frac{\partial y_t^T}{\partial t} = \frac{\partial A_t^T}{\partial t} + \left[\frac{\partial \left(\vec{B}_t^T \right)'}{\partial t} \vec{x}_t \right]. \quad (48)$$

However, because of the time-homogeneity assumption,

$$\frac{\partial y_t^{T_i}}{\partial t} = - \frac{\partial y_t^{T_i}}{\partial T} \quad (49)$$

and therefore

$$\frac{1}{P_t^T} \frac{\partial P_t^T}{\partial t} = y_t^T + (T - t) \frac{\partial y_t^T}{\partial T}. \quad (50)$$

Looking at equation (44) for the total portfolio convexity, $\mathfrak{CP}$, and substituting the expression we have just obtained for $\frac{1}{P_t^T} \frac{\partial P_t^T}{\partial t}$, we get

$$\begin{aligned} \mathfrak{CP} &= \frac{1}{2} \text{Tr}[S' DS] \\ &= - \sum_{i=1,n} \omega_i \left[\left(y_t^{T_i} - r_t \right) + (T_i - t) \frac{\partial y_t^{T_i}}{\partial T_i} \right], \end{aligned} \quad (51)$$

where $\frac{\partial y_t^{T_i}}{\partial T_i}$ designates the slope of the yield curve evaluated at maturity T_i .

The two terms on the right-hand side have well-known market interpretations: the quantity $(y_t^{T_i} - r_t)$ tells the investor how much a given bond ‘carries’ (i.e. how much extra yield it returns above the funding cost, r_t)—and therefore the term $\sum_{i=1,n} \omega_i (y_t^{T_i} - r_t)$ tells the trader how much the whole portfolio ‘carries’.

The term $\frac{\partial y_t^{T_i}}{\partial T_i}$ is also part of the market lore: it is the ‘roll-down’, i.e. the change in yield as the maturity changes by a small amount, multiplied by the duration of the bond.[†]

Whatever their names and interpretations, the important observation is that also the quantities $(y_t^{T_i} - r_t)$ and $\frac{\partial y_t^{T_i}}{\partial T_i}$ can be read from the shape of the market yield curve and are therefore model-independent.

Putting the pieces together we therefore have

$$\mathfrak{CP}_{theor} = \frac{1}{2} \text{Tr}[S' DS] = - [Carry + Roll] \quad (52)$$

where

$$Carry = \sum_{i=1,n} \omega_i Carry_i \quad (53)$$

$$Roll = \sum_{i=1,n} \omega_i Roll_i \quad (54)$$

and

$$Carry_i = y_t^{T_i} - r_t \quad (55)$$

$$Roll_i = (T_i - t) \frac{\partial y_t^{T_i}}{\partial T}. \quad (56)$$

In words: the portfolio convexity is equal to the opposite of the sum of the portfolio carry and roll-down. Note that we have appended the subscript ‘*theor*’ to the symbol for the portfolio convexity, $\mathfrak{CP}_{theor}$, in order to emphasize that this is what is convexity *should* be, given the shape of the yield curve.

9. Linking the term $\frac{1}{2} \text{Tr}[S' DS]$ to yield volatilities

We now want to link the convexity term $\frac{1}{2} \text{Tr}[S^T DS]$ to the volatilities of yields. To do so, recall the definition of the matrix D ,

$$D = \sum_{i=1,n} \omega_i \left(B_t^{T_i} \right) \left(B_t^{T_i} \right)'.$$

Since we know that

$$y_i = \alpha_t^{T_i} + \left(b_t^{T_i} \right)^T x_t \quad (57)$$

with

$$b_t^{T_i} = - \frac{1}{T_i - t} B_t^{T_i} \quad (58)$$

it follows from equations (15) and (16) that

$$\frac{1}{2} \text{Tr}[S' DS] = \frac{1}{2} \sum w_i (T_i - t)^2 \sigma_{y_i}^2. \quad (59)$$

So, now we have equated the term $\frac{1}{2} \text{Tr}[S^T DS]$ to a simple function of the yield volatilities. Combining this with the previous results, we have:

$$\mathfrak{CP}_{theor} = \frac{1}{2} \text{Tr}[S' DS] = - [Carry + Roll] \quad (60)$$

and therefore

$$\begin{aligned} &\frac{1}{2} \sum w_i (T_i - t)^2 \sigma_{y_i}^2 \\ &= - \sum_{i=1,n} \omega_i \left[\left(y_t^{T_i} - r_t \right) + \left((T_i - t) \frac{\partial y_t^{T_i}}{\partial T} \right) \right]. \end{aligned} \quad (61)$$

The important observation is that, apart from the weights, all the quantities in the expression above can be either read from the shape of the market yield curve—the quantities on the right-hand side—or empirically calculated—the yield volatilities on the left-hand side. Again apart from the weights, ω_i , *there are no model parameters left*.

The equation just obtained is an ‘equilibrium’ relationship, which tells us how the yields in our economy *should* behave. To derive it we placed ourselves in a deterministic-volatility affine framework, and obtained the quantities that appear in equation (61). The unobservable model parameters (such as its reversion-speed matrix, or the volatility of the unspecified factors) have disappeared from the equation. We don’t even

[†]Recall that the weight ω_i ‘contains’ the bond price, and that the duration of a discount bond is given by $\frac{dP_t^T}{dy_t^T} = -P_t^T (T - t)$.

have to specify what the state variables are, or how many state variables the model has, as long as we can assume that the market yield curve and the covariance matrix have been recovered accurately during the calibration phase. So, the only condition the approach presented so far requires is the existence of *some* affine model that can recover the market yield curve and the covariance matrix sufficiently well. As we said, we know that Principal-Component-based affine models, such as the one in [Rebonato et al. \(2014\)](#), enjoy these properties. This is all we need for the treatment we present below.

10. Making the weights (almost) model-independent

So far, the immunizing weights are, of course, model dependent because the weights ω_i depend on the ‘factor durations’, $\frac{\partial P_t^{T_i}}{\partial x_k}$. If we want to obtain these from the chosen affine model, they will depend on the reversion speeds (although it can easily be shown that they do not depend on the factor volatilities). However, suppose that we choose as our factors principal components. Then we have

$$d\vec{y} = \underline{V} d\vec{x}, \quad (62)$$

where V is the matrix of eigenvectors of the covariance matrix of yield changes.

From the affine relationship we also know that

$$d\vec{y} = (\underline{b})' d\vec{x}_t, \quad (63)$$

where the matrix $(\underline{b})^T$ is obtained by stacking the row vectors, $(\vec{b})^T$. Therefore, if the state variables are principal components, one must have

$$\underline{V} = (\underline{b})'. \quad (64)$$

This means that, thanks to equation (62), we can obtain the derivatives of the yields with respect to the factors (that enter the matrix $(\underline{b})^T$) simply by looking up the appropriate element of the matrix V , because the derivative of the i -maturity yield with respect to the k th principal-component factor, $\frac{\partial y_t^{T_i}}{\partial x_k}$, is just given by the i, k element of the V matrix:

$$\frac{\partial y_t^{T_i}}{\partial x_k} = (V)_{ik}. \quad (65)$$

The quantity $\frac{\partial P_t^{T_i}}{\partial x_k}$ is then easily obtained as

$$\frac{\partial P_t^{T_i}}{\partial x_k} = \frac{\partial P_t^{T_i}}{\partial y_t^{T_i}} \frac{\partial y_t^{T_i}}{\partial x_k} = -(T_i - t) P_t^{T_i} (V)_{ik}. \quad (66)$$

Summarizing: from

$$\sum_{i=1,n} \left(\overrightarrow{\omega_i \frac{1}{P_t^{T_i}} \frac{\partial P_t^{T_i}}{\partial x_k}} \right) = \vec{0}. \quad (67)$$

and

$$\frac{\partial P_t^{T_i}}{\partial x_k} = -(T_i - t) P_t^{T_i} (V)_{ik} \quad (68)$$

we have

$$\sum_{i=1,n} \left(\overrightarrow{\omega_i (T_i - t) (V)_{ik}} \right) = \vec{0} \quad (69)$$

Without loss of generality we can set $\omega_1 = 1$, and therefore

$$\begin{aligned} & \begin{bmatrix} \omega_2 \tau_2 (V)_{12}' \\ \omega_2 \tau_2 (V)_{22}' \\ \dots \\ \omega_2 \tau_2 (V)_{n-1,2}' \end{bmatrix} + \dots + \begin{bmatrix} \omega_n \tau_n (V)_{1n}' \\ \omega_n \tau_n (V)_{2n}' \\ \dots \\ \omega_n \tau_n (V)_{n-1,n}' \end{bmatrix} \\ &= \begin{bmatrix} -\tau_1 (V)_{11}' \\ -\tau_1 (V)_{21}' \\ \dots \\ -\tau_1 (V)_{n-1,1}' \end{bmatrix}, \end{aligned} \quad (70)$$

with $\tau_i \equiv T_i - t$.

The duration-neutralizing weights can, in theory,[†] be obtained by a matrix inversion:

$$\vec{\omega} = A^{-1} \vec{q}. \quad (71)$$

These are the statistically determined weights we were looking for.

We stress the two levels of assumptions enforced so far.

Up to the determination of the weights, we only required the setting to be that of a deterministic-volatility time-homogeneous affine model capable of recovering well the yield curve and the covariance matrix. Any affine model that produces these results will do.

For the determination of the weights we have then required the affine model to use principal components as its state variables, and we use the equivalence results in section 2. We have used the empirical eigenvectors to determine the weights that immunize the portfolio. This is not an inconsistency, because a principal-component-based affine model can always be constructed to have exactly these empirical eigenvectors as the corresponding ‘durations’.

11. The set-up of the empirical investigation

We now explore empirically the predictive power of the approach outlined above.

11.1. The data

We used discount bond prices for our analysis. Discount bonds are not (actively) traded, but discount bond prices closely consistent with the prices of traded coupon-bearing Treasury bonds have been obtained and kindly made publicly available by [Gurkaynak et al. \(2007\)](#).[‡] These data stretch back to the 1960s, but for the convexity study of interest here we must concentrate on long-dated bonds, for which reliable prices only became available after March 1985. Therefore we present results based on the period 1985 to 2015.

[†]We say ‘in theory’ because, as we show below, this is in practice not the most efficient way to obtain them. See section 10.3 in this regard.

[‡]These discount prices were obtained using the [Nelson and Siegel \(1987\)](#) method, as described in detail in [Gurkaynak et al. \(2007\)](#). The Authors there describe the (high) accuracy of price recovery achieved by the model.

11.2. The trading strategy

During this period, each week we go long or short convexity depending on whether the right-hand side of equation (61) is larger or smaller, respectively, than the left-hand side, and making sure that the portfolio is immunized against the factors. We describe below how exactly the immunization is carried out. We hold the position for a week, at the end of which period we unwind the portfolio, we record the profit or loss, and we establish a new position in the same manner. We do not take transaction costs into account. The cumulative profit or loss (always ‘profit’ for short in what follows) and all the other statistics reported below refer to being long or short in the longest-maturity (30-year) bond exactly one unit of currency.

11.3. Determining the weights in practice

In equation (71) we showed how the optimal weights in the hedged portfolio can be obtained as the solution of a linear problem, i.e. from a matrix inversion. As it is often the case, when we perform this matrix inversion we often find highly leveraged solutions with very large and positive weights on some bonds offset by similarly large and negative weights on other bonds. These leveraged portfolios are not just impractical to put in place; more seriously, these very positive and very negative positions act as noise-and-small-errors-amplifiers, making the detection of the true value of convexity all but impossible.

To circumvent this problem, we proceed as follows. Given a number of bonds, (typically three, but sometimes four[†]), we determine at time t the portfolio weights that minimizes the portfolio variance given the actual yield curve moves experienced in the N days before time t . These minimum-variance weights are obtained by a simple, low-dimensionality non-linear search. The number of days, N , was chosen in our study to be 300 business days, and the optimal weights—and, more importantly, the results of the strategy—turned out to be largely insensitive to the exact choice of N .[‡] Most importantly, as we show in what follows, the resulting weights turned out to be much more stable than the matrix-inversion weights.

11.4. Estimating the yields volatilities

The other crucial input for the strategy are the predicted volatilities of the yields. We used one of the simplest volatility estimators of the GARCH family, namely the GARCH(1,1) model: the prediction of the variance, σ_i^2 , of the a yield, y , at time $i \Delta t$ for time $(i + 1) \Delta t$ is given by

$$\sigma_i^2 = a\sigma_{i-1}^2 + b \left(\Delta y_{i-1}^2 \right) + c\hat{\sigma}^2 + \epsilon_i, \quad (72)$$

[†]The results were qualitatively very similar when we used four- or three-bond portfolios. We report in the following the results from deploying three-bond portfolios only, but the results for four-bond portfolios did not differ materially.

[‡]An alternative approach would be the use of the technique of Singular Value Decomposition. In our tests we found little difference in profit-and-loss performance between the two approaches. However, the numerical procedure described in this section gave rise to more intuitive and somewhat more stable weights, and therefore we adopted it in the rest of the study.

with the usual constraints on the $\{a, b, c\}$ coefficients.

We found that equation (72) provides a good trade-off of simplicity, stability, transparency and responsiveness. As the immense literature on the topic testifies, there are far more powerful predictive techniques. As mentioned in the introduction, however, we did not employ on purpose a sophisticated volatility estimate, because we wanted to test whether excess returns could be made using a reasonably reactive, but otherwise very simple, estimate of future volatility. We are not testing, in other words, whether we have an especially good volatility crystal ball, but whether the market prices convexity correctly, given a robust, simple, *and hence presumably commonly-available*, predictive estimate of future volatility. (In any case, Hansen and Lunde (2004) carry out an out-of-sample comparison of 330 different volatility models applied to foreign exchange and stoch prices, and find that ‘the best models do not provide a significantly better forecast than the GARCH(1,1) model.’)

In sum: all the results that we report in what follows were obtained using a GARCH estimate of volatility as per equation (72), and with the weights derived from the non-linear, minimum-variance, 300-day optimization described in the previous subsection.

12. Empirical results

12.1. Is the yield curve fairly curved?

In the first part of this section we report the results from engaging over a 30-year period in three typical convexity strategies, namely the strategies based on

- a portfolio made up of 5-year, 10-year and 30-year bonds (referred to as Portfolio 5/10/30 in what follows);
- a portfolio made up of 10-year, 20-year and 30-year bonds (referred to as Portfolio 10/20/30 in what follows);
- a portfolio made up of 5-year, 15-year and 30-year bonds (referred to as Portfolio 5/15/30 in what follows).

The cumulative profits made from the 5-10-30-year strategy is shown in figure 6. Figure 7 then shows the weights of the bonds of maturity other than 30 years for the same portfolio. For all the strategies the holding in the 30-year bond was always normalized to 1. Similar results were obtained for the other strategies. The accompanying figures are not shown, but the results are reported in the tables below.

It is important to pause briefly to explain what is shown in figure 7. The two lines show the amounts of 5-year and 10-year bonds needed to insulate the portfolio against yield-curve moves, *conditional on the position in the 30-year bond being equal to +1*. We note that during the period under analysis this duration-neutral portfolio is sometimes obtained by being long the 5-year and 30-year bonds and short the 10-year bond, and sometimes by being long the middle and long bonds and (very) short the short-maturity bond. Which combination is preferred depends on how the ‘level’ and ‘slope’ exposures are immunized against by the minimum-variance algorithm.

Keeping this in mind, the first observation is that the weights are well-behaved, and their magnitude intuitively understandable. So, for instance, the magnitude of the weight in the 5-year bond tends to be large because of the short duration of the bond: for a given shock, ‘more of’ the 5-year bond is needed to duration-hedge the corresponding movement in the 30-year bond. It is also clear that the changes in the weights of the 5- and 10-year bonds tend to be strongly anticorrelated.

It is interesting to examine the sign of the suggested strategy. We can do so by introducing a convexity indicator, defined as +1 when the historical volatility is greater than the ‘implied’ volatility (i.e. when we should be ‘long’ convexity), and –1 when the reverse occurs. The figures 8 and 9 show the convexity indicator in the post crisis period, and the rolling 3-week average of the convexity indicator for the whole period. The whole-sample average of the convexity indicator turned out to be 0.283, indicating that, on average, the yield curve was not ‘enough curved’.

The last observation is that the three strategies are all overall profitable, with a Sharpe Ratio for the whole sample ranging between 0.4 and 0.5. For the strong-signal, post-2008 part of the sample the Sharpe Ratios are considerably higher, ranging from 0.9 to 1.4. See table 3.

12.2. Why are the strategies not always profitable?

The strategy described above could fail to be profitable because the yield curve curvature was ‘fair’. However, if our approach were correct (and neglecting transaction costs), the strategy described above should never lose money. Figure 6 shows that, empirically, this was clearly not the case.

In order to understand the reason(s) for the profitability of the strategies, we therefore begin by looking at the strength of our signal – where we define as signal the difference between the convexity projected using our volatility predictions (the term $\frac{1}{2} \sum w_i (T_i - t)^2 \sigma_{y_i}^2$), and the convexity ‘embedded’ in the curvature of the yield curve (the term $-\sum_{i=1,n} \omega_i \left[\left(y_i^{T_i} - r_t \right) + \left((T_i - t) \frac{\partial y_i^{T_i}}{\partial T} \right) \right]$).

If we make money when the signal is strong, and we fail to make money when it is weak, it would be *prima facie* evidence that we are indeed capturing phases of different ‘curvature fairness’; if the strength of the signal failed to have much of an explanatory power, it would suggest that either we are not guessing correctly the future volatility, or we are not properly immunized.

Figure 10 shows the cumulative profit from the three strategies together with the absolute value of the signal. In the same picture we also report the cumulative profits obtained following two slightly more robust strategies: in the curve labelled ‘Select’ we show the combined cumulative profit made by trading only when the three strategies are all concordant (i.e. they all suggest to be either long or short convexity); in the curve labelled ‘Combined’ we show the cumulative profit made by adding up the three strategies. In all cases we expect a large and positive slope when the absolute value of the signal is large.

The two panels convey a complex message. Looking at the more recent (post 2008) period, it is clear that there is indeed a positive correlation (quantified in the next section) between the profitability of the strategy (and the Sharpe Ratios reported in table 4) and the strength of the signal. The failure to make money in the 10-year period than ends in 2008 can also be explained by looking at the weakness of the signal. It is the earliest period, when the signal is strongest, yet relatively little money is made (and sometimes money is lost) that requires some explaining.

Recall that there can be three reasons for not making money at every time step.

- (1) The yield curve is fairly curved—if this is why we don’t make money, then the strength of the signal should be small when the cumulative profit remains flat. We can test this.
- (2) We are not really duration-immunized with respect to the underlying factors. In this case we have an unwanted residual exposure to the level, slope or curvature principal components. We can test this by regressing the daily profit against changes in the three first Principal Components.
- (3) We are bad at predicting the future yield volatilities. We can test this as well: if this is why we do not make money, we would get a significant R^2 from regressing the convexity-attributable profit (which we call in the following the ‘stripped profit’) against the difference between the predicted and the realized convexity.

12.3. Is the strength of the signal correlated with the profitability of the strategy?

We start by looking at whether the absence of profit on a particular trading period is simply due to the fact that there is no money to be made, i.e. to the fact that curve is fairly curved. If this were the case we would observe a strong correlation between the profit actually made and absolute value of the signal strength (the absolute value, because we don’t care whether the signal is positive or negative, as long as it is loud and clear). To test this we sorted the realized (signed) profit by the strength of the absolute signal. Then we took the average of the profit in each decile, and we plotted it in figure 11. If there were a clear relationship between signal strength and profit made, we would expect a graph monotonically increasing with the decile.

The picture is a bit more complex. As figure 11 shows, when the signal is weak, indeed, we make very little money (see the first three deciles). Also, the last decile has the largest average profit, and the average profit in last two deciles are in the top three averages. This ‘impressionistic’ depiction of the dependence can be made more precise by regressing the average profit in each decile against the decile. The results of this regression are shown in figure 12, which also displays the trend line and an R^2 of almost 30%.

Clearly, the signal is informative, but there is a lot more than signal strength contributing to the realized profit. We try to understand what this could be in the next two sub-sections.

Table 3. The Sharpe Ratios for the three strategies, over the whole sample (second column), over the first and second halves of the sample (third and fourth column) and over the post-2008, strong-signal period (right-hand-side column).

	SR (whole)	SR (1st half)	SR (2nd half)	SR (post-2008)
5-10-30	0.58	0.52	0.63	1.21
10-20-30	0.45	0.32	0.58	0.93
5-15-30	0.61	0.52	0.70	1.41

12.4. Explaining the residuals—residual duration exposure?

The first candidate explanation for the observed lack of strong correlation between signal strength and profit made is the possible failure to immunize against the duration exposures to the first principal components. If this were indeed the case we would observe significant coefficients in the regression of profit changes against changes in the first, second and third principal components (dubbed level, slope and curvature in the following) of the three-bond yield curve of interest.

What we find is that the slope coefficients, reported in tables 4–6 are rarely significant, and the R^2 for the more volatile regressor (i.e. the changes in level) is non-significant. There is a relatively high R^2 coefficient (27.26%) and significant slope and intercept coefficients for the 10-20-30 portfolio, but the largest R^2 corresponds to the least volatile of the three factors, i.e. the curvature.

The same message is clearly shown in figure 13, which displays the cumulative profit from the 5-10-30 strategy and the cumulative profit explained by the residual duration exposure—clearly, the cumulative profit from the residual duration exposure is almost two orders of magnitude smaller than the cumulative profit from the strategy.

Similar results (not shown) were found for the other two strategies.

Overall, one can clearly say that our minimum-variance-portfolio did a good job at neutralizing the portfolio against first-order (‘duration’) exposures, and that the occasional periods of negative returns are not mainly due to a residual level or slope exposure. We therefore explore in the next sub-section the last possible source of error, i.e. the poor prediction of the future volatility.

12.5. Explaining the residuals—wrong volatility estimate?

The second possible explanation for the difference between the profit actually made and the prediction of the profit that we ‘should’ make is our failure to guess the realized volatility correctly. Was this the case?

To test this possibility we took the daily realized profit or loss made by deploying our quasi-agnostic convexity strategy, and we stripped out whatever first-order contributions there may be from the changes in yields.[†] The reasoning then goes as follows. First we regress the stripped profit (as defined

above) against the difference between the theoretical portfolio convexity and the realized convexity.

The theoretical convexity is given by $\mathcal{CP}_{theor} = -[Carry + Roll]$ —it is the model-independent quantity (the ‘implied convexity’) that can be read off the yield curve. If the yield curve were ‘fairly’ curved, we should have $\mathcal{CP}_{theor} = \frac{1}{2} \sum \omega_i \hat{\sigma}_i^2 T_i^2$. We can regard this as the definition of a set of ‘implied’ volatilities, $\hat{\sigma}_i$. The realized convexity is given by $\frac{1}{2} \sum \omega_i \Delta y_i^2 T_i^2$, where Δy_i is the observed change in the i th yield. Therefore, as a first step, we regress the stripped profit against $\frac{1}{2} \sum \omega_i (\Delta y_i^2 - \hat{\sigma}_i^2) T_i^2$.

When we carry out the regression of the stripped profit against the difference between the implied and realized convexity, for all portfolios we get an R^2 statistically indistinguishable from 1, and a slope and intercept statistically indistinguishable from one and zero, respectively. This, of course, tells nothing more than we have stripped our profits and losses accurately (that we have ‘delta-hedged’ well).

With this reassuring result about the effectiveness of our stripping procedure, we now consider the differences between the implied and realized convexity. More precisely we regress the stripped profit against the difference between the *realized* convexity, $\frac{1}{2} \sum \omega_i (\Delta y_i^2 - \hat{\sigma}_i^2) T_i^2$, and the *predicted* convexity, $\frac{1}{2} \sum \omega_i (\Delta y_i^2 - \tilde{\sigma}_i^2) T_i^2$, and look at the deterioration of the R^2 from the value of almost 1 that we obtained before. Any reduction in R^2 , or any statistically significant deviations of the intercept and slope from (0,1) would indicate how much of the deviation of the stripped cumulative profit from the signal strength is due to our volatility misestimation.

The results of these regressions are shown in table 7 for the 5/15/30 strategy.

The regression coefficients are all very significant and close to (0,1), but the R^2 statistic shows a dramatic reduction from the value of 1 we obtained before. This—the difference between $\hat{\sigma}_i^2$ and $\tilde{\sigma}_i^2$ —is therefore the main reason for our failure to make (or, at least, not lose) money over every rebalancing period.

This is, after all, not surprising. By examining carefully the time series, it becomes apparent that most of the times there is a strong positive relationship between the difference in convexities and the stripped profit. However, this relationship is occasionally spoiled when large unexpected moves occur. Such ‘surprises’ are by definition unpredictable, and it is in these situations that our projected variances fail to match the implied ones.

So, the picture that is conveyed by this part of the analysis is the following. Usually, one can predict fairly accurately the volatility that will be realized over the next time step. In these ‘normal’ market conditions the model-agnostic convexity strategy as implemented has good predictive power—by which

[†]The stripped P&L, $SP\&L_i$, for bond i was calculated as $SP\&L_i = Dur_i \Delta y_i$, where Dur_i is the duration of the T_i -maturity bond. The stripped P&L for the portfolio, $SP\&L$, was then calculated as $SP\&L = \sum_i \omega_i SP\&L_i = \sum_i \omega_i Dur_i \Delta y_i$.

Table 4. The R^2 statistic, the intercept (a) and the slope (b) of the regression, with the standard error of the estimates in parenthesis, for the regression of the profit changes against changes in level for the three strategies.

<i>Level</i>	R^2	a	b
Port 5/10/30	0.429%	0.0017(0.0007)	1.2519(0.5671)
Port 10/20/30	1.044%	0.0011(0.0006)	-1.7764(0.5144)
Port 5/15/30	0.011%	0.0015(0.0006)	0.1989(0.5574)

Table 5. Same as table 4 for changes in slope.

<i>Slope</i>	R^2	a	b
Port 5/10/30	14.43%	0.0015(0.0006)	8.3794(0.6069)
Port 10/20/30	10.77%	0.0011(0.0006)	-1.3264(0.5144)
Port 5/15/30	3.639%	0.0014(0.0007)	3.9990(0.6121)

Table 6. Same as table 4 for changes in curvature.

<i>Curvature</i>	R^2	a	b
Port 5/10/30	13.61%	0.0016(0.0007)	8.6202(0.6461)
Port 10/20/30	27.26%	0.0011(0.0005)	-7.6940(0.6589)
Port 5/15/30	0.541%	0.0015(0.0006)	1.3552(0.5468)

Table 7. The R^2 statistic, the intercept (a) and the slope (b) of the regression, with the standard error of the estimates in parenthesis, for the regression of the ‘stripped’ profit changes against the difference between the predicted and realized portfolio convexity.

	R^2	a	b
<i>Port 5/10/30</i>	17.38%	0.0000(0.0000)	0.9687(0.0629)
<i>Port 10/20/30</i>	16.07%	-0.0001(0.0000)	1.0970(0.0756)
<i>Port 5/15/30</i>	18.69%	0.0000(0.0000)	1.0120(0.0628)

we mean that it identifies correctly whether the yield curve is fairly curved or not. If it is not, in these conditions, the strategy is generally profitable.

When exceptional (and hence unpredictable) moves do occur, the strategy ceases to work as predicted—it may fail, or it may ‘work too well’, depending on the size of the move, and on whether the strategy happened to be long or short convexity before the unexpected-size move.

We note that there is an intrinsic asymmetry (skewness) to the profit profile from the strategy. Consider the case first when the investor is long convexity before the move. A small loss will be made if the realized move is ‘too small’ (a ‘theta-decay loss’ from an option perspective), and an unexpectedly large gain will be made if the move is exceptionally large: if the investor is *long* convexity, the maximum loss that can be incurred is the ‘theta decay’ over the time step (which will occur for a zero move in yields); the gain is unlimited with the size of the move. The reverse occurs if the strategy is short convexity. Therefore the largest unexpected losses correspond to situations when the investor was short convexity and unexpectedly large moves occurred.

13. Conclusions

We have tested whether the US Treasury market has priced convexity correctly in the last 30 years. We have worked in the broad framework of affine modelling, but we have made our results as specific-model-independent as possible. We have shown that the theoretical profitability of a convexity strategy (which we decompose as the sum of several financially interpretable terms) is purely a function of the shape of the yield curve, and of the yield volatilities.

Thanks to a simple equivalence result proven in section 2, we show that all affine models that recover exactly the shape of the yield curve and the yield volatilities will provide exactly the same predictions of what the theoretical value of convexity should be. So, we use a principal-component-based affine model in our approach, but we know that our results can be generalized to a wide class of affine models.

We have shown that, when cast in these terms, a ‘convexity trade’ is akin to the practice of ‘gamma trading’ in the option domain, with the curvature of the yield curve (and the associated ‘yield give-up’ incurred to establish a long-convexity

strategy) playing the role of the premium paid to be long convexity. Exactly as in option trading, in equilibrium (i.e. if the theoretical curvature matches the true curvature of the term structure) the investor will recover the 'premium' by the profit made during the delta-re-hedging part of the strategy.

We find that the strategy based on the signal we present in the study has a Sharpe Ratio ranging between 0.45 and 0.61 for the full-sample study. In the post-2008 period the Sharpe Ratio is significantly higher. This suggests that

- (1) there are extended periods when the curvature of the yield curve is not 'correctly priced' in the sense that conditional strategies with attractive Sharpe Ratios can be put in place; and
- (2) that it is possible to identify *ex ante* both when this is the case, and the sign of a profitable strategy.

Of course, our analysis could be perfectly correct, and yet the strategy could make no money at all simply because convexity is fairly priced (i.e. the term structure of yields is always 'correctly' curved). It is therefore very interesting that we find a statistically significant positive relationship between the sub-period Sharpe Ratios (or the slope of the cumulative profit curve) and the strength of our signal. The R^2 of the regression of predicted against realized deciles is not as high as we may have hoped, but the correlation is statistically significant.

If our model were perfect, and our prediction of volatility infallible, we could not lose money. This is not the case. Residual first-order exposure to the first three principal components explains little of the unexpected profit. Most of the discrepancies between the predicted and the realized profits come from our failure to predict correctly the instantaneous realized volatility over the next step. We have identified unexpectedly large moves in yields when the investor is short convexity as one of the greatest contributors to these negative 'surprises'.

Disclosure statement

No potential conflict of interest was reported by the authors.

References

- Adrian, T., Crump, R.K. and Moench, E., Pricing the term structure with linear regressions. Federal Reserve Bank of New York, Staff Report No 340, August 2008. Revised 2013, 2014. Available online at: http://www.newyorkfed.org/research/staff_reports/sr340.pdf (last accessed 15 October 2014).
- Brown, R.H. and Schaefer, S.M., Why long term forward interest rates (almost) always slope downwards. London Business School working paper, 2000. <http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.196.7176&rep=rep1&type=pdf> (last accessed 26 January 2015).
- Dai, Q. and Singleton, K.J., Specification analysis of affine term structure models. *J. Finance*, 2000, **55**(5), 1943–1978.
- Deguillaume, N., Rebonato, R. and Pogudin, A., The nature of the dependence of the magnitude of rate moves on the rates levels: A universal relationship. *Quant. Finance*, 2013, **13**(3), 351–367.
- Diebold, F.X. and Rudebusch, G.D., *Yield Curve Modelling and Forecasting—The Dynamic Nelson-Siegel Approach*, 2013 (Princeton University Press: Princeton, NJ).
- Fontaine, J.-S. and Garcia, R., Bond liquidity premia. *Rev. Financ. Stud.*, 2012, **25**(4), 1207–1254.
- Gurkaynak, R.S., Sack, B. and Wright, J.H., The US treasury yield curve: 1961 to the present. *J. Monetary Econ.*, 2007, **54**, 2291–2304. Available online at: <http://www.federalreserve.gov/econresdata/researchdata/feds200805.xls> (last accessed 27 January 2015).
- Hansen, P. and Lunde, A., A forecast comparison of volatility models: Does anything beat a GARCH(1,1) Model?. *J. Appl. Econom.*, 2004, **20**, 873–889. Available online at: <http://www.federalreserve.gov/pubs/feds/2006/200628/200628abs.html> (last accessed 20 September 2014).
- Heath, D., Jarrow, R.A. and Merton, R., Bond Pricing and the term structure of interest rates: A new methodology. Working Paper (revised edition), Cornell University, 1989.
- Joslin, S., Priebsch, M. and Singleton, K.J., Risk premiums in dynamic term structure models with unspanned macro risks. *J. Finance*, 2014, **LXIX**(3), 1197–1233.
- Ludvigson, S.C. and Ng, S., Macro factors in bond risk premia. *Rev. Financ. Stud.*, 2009, **22**, 5027–5067. Available online at: <http://rfs.oxfordjournals.org/content/22/12/5027.abstract> (last accessed 6 May 2015).
- Nelson, C.R. and Siegel, A.F., Parsimonious modeling of yield curves. *J. Bus.*, 1987, **60**, 473–489.
- Joslin, S., Singleton, K.J. and Zhu, H., A new perspective on gaussian dynamic term structure models. *Rev. Financ. Stud.*, 2011, **24**, 926–970.
- Rebonato, R., Saroka, I. and Putyatin, V., A Principal-component-based affine term structure model. Working Paper. submitted for publication in *Applied Mathematical Finance*. 2014. Available online at: http://papers.ssrn.com/sol3/papers.cfm?abstract_id=2451130.
- Wu, J.C. and Xia, F.D., Measuring the macroeconomic impact of monetary policy at the zero lower bound. Working Paper Chicago Booth and NBER, 2014. Available online at: <http://faculty.chicagobooth.edu/jing.wu/research/pdf/wx.pdf> (last accessed 31 March 2015).

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